Contents

Active Credit-Portfolio Optimization: Integrating Machine-Learning Underwriting, Endogenous Stiglitz-Weiss Risk Pricing, and Basel III/IV Economic Capital **1**

Abstract . . . 1
1. Introduction and Research Question . . . 2
2. Theoretical Framework and Literature Review . . . 6
3. Data and Empirical Estimation . . . 9
4. Methodology and Algorithmic Pipeline . . . 11
5. Empirical Results and Regulatory Risk Analysis . . . 20
6. Managerial Implications and Policy Discussion . . . 41
7. Conclusion, Limitations, and Future Research . . . 43
References . . . 45
Appendix A. Repeated Random Splits (seeds 0–19) . . . 47
Appendix B. Base-Case Parameters . . . 48
Appendix C. Notation . . . 49
Appendix D. Proof of Proposition 1 . . . 49
Appendix E. Reproducibility . . . 50
List of Figures . . . 51

Active Credit-Portfolio Optimization: Integrating Machine-Learning Underwriting, Endogenous Stiglitz-Weiss Risk Pricing, and Basel III/IV Economic Capital

Author: Niv Vaisman Zancho
Degree: Master of Arts in Financial Economics, Reichman University
Supervisor: Prof. Doron Avramov
Date: September 2026
JEL Classification: G21, G32, C53, C61, D82
Keywords: credit rationing, adverse selection, economic capital, RAROC, Vasicek ASRF, machine learning, probability calibration, mechanism design, contract pricing, net present value

---

Abstract

Lenders increasingly score applicants with machine-learning models, yet most still use the score only to accept or reject a contract whose terms were set elsewhere. This thesis develops a model-agnostic decision layer that instead co-optimizes principal, maturity and rate, customer acquisition cost and behavioral acceptance to maximize risk-adjusted return on capital, and evaluates it against nine passive underwriting regimes on a public marketplace-lending testbed of 10,000 loans, strictly out of sample. Default responds to contract terms

through empirically estimated Stiglitz–Weiss channels ($\widehat{\gamma}_{emp} = 0.0898$ per rate point, $\widehat{\delta}_{emp} = 6.575$ per unit of payment-to-income). The engine prices each of 1,178 candidate contracts per borrower under an explicit loss profile, a calibrated probability of default, heterogeneous and partially observed demand with Bayesian updating after a rejection, a Basel ASRF capital charge inside the objective, and deployability constraints, and routes offers through a three-tier bargaining tree gated on its expected value. An inverse semi-elasticity pricing rule characterizes the optimum. In the base case the engine earns an expected net profit of **$12.55M** against **$8.62M** for the best passive regime (**1.456×**; paired-bootstrap 95% interval 1.40–1.51), with a lifetime RAROC of **99.9%** against 79.6% at a comparable approval rate. A pseudo-realized backtest that re-prices default gives 1.23×. The advantage survives six loss profiles (1.35–1.51), fourteen ±25% parameter shocks, a ten-fold increase in demand elasticity (1.05), zero acquisition cost (1.40), an alternative PD model (1.43), and twenty re-estimated random partitions, all of which the engine wins. It is almost entirely a pricing advantage: the rate axis delivers 94% of the gain, principal cuts the rest and maturity nothing. It concentrates in the middle of the credit distribution rather than on the riskiest or lowest-income borrowers. It is conservative with respect to the rate elasticity of default, which is not identified within credit grades. The results support a restated *Machine-Learning Underwriting Paradox*: discrimination does not order value, since a model with a statistically superior AUC earns less, and a classifier used only as a filter leaves roughly a third of the attainable value unexploited, because calibration and pricing, not ranking, drive profit.

---

1. Introduction and Research Question

1.1 Motivation: The Structural Gap in Passive Machine-Learning Credit Scoring

Over the past decade credit underwriting has migrated from expert-judgment rating grids toward supervised classifiers that estimate a borrower's probability of default (PD) with high discriminatory power. The implicit premise of that migration is that a better *risk ranking* mechanically produces better *lending decisions*. This thesis argues that the premise is incomplete in a way that is economically material and empirically measurable.

A risk classifier answers one question, *how likely is this borrower to default?*, and converts the answer into an accept/reject screen on a contract whose terms were set elsewhere. A lender's objective is different: shareholder value per unit of scarce regulatory capital. That objective depends jointly on the *price* charged, the *structure* of the contract (principal and maturity), the *cost of acquiring* the borrower, the *capital* the exposure consumes, and, because of Stiglitz and Weiss (1981), on the fact that the borrower's default probability responds to the first two. A model that optimizes the ranking while holding all of these fixed optimizes a proxy, not the objective. Hand (2009) makes the statistical version of this point: the AUC summarizes a classifier over all possible cost thresholds, most of which no lender would ever use; Verbraken, Bravo, Weber, and Baesens (2014) and Serrano-Cinca and Gutiérrez-Nieto (2016) make the financial version, showing that profit-based evaluation of consumer-credit models re-orders the models that accuracy-based evaluation ranks.

The engine developed here is intended as a general decision layer for lenders that already use, or are adopting,

statistical and machine-learning risk models: banks, specialized consumer lenders and fintech platforms. It is deliberately model-agnostic. Any PD model, from a bureau scorecard to a gradient-boosted ensemble, enters only through a calibrated probability, and the remaining inputs (loan terms, income, the lender's cost of funds, capital hurdle, recovery experience and acquisition costs) are quantities every regulated lender already measures. The LendingClub data used in this thesis are therefore a public, well-studied *testbed* for the framework, not its intended domain; Section 6.1 lists what a lender would need to supply to apply it to its own book.

1.2 The Machine-Learning Underwriting Paradox, Restated

We call the gap between ranking quality and value creation the **Machine-Learning Underwriting Paradox**. The proposal conjectured its strong form: a high-AUC classifier, used passively, could *destroy* value relative to a legacy scorecard. The evidence in this thesis supports a more careful statement. Under an immediate-loss, zero-recovery loss profile (the specification of the earlier prototype, retained here as a conservative bound), passive regimes do post negative *realized* profit; under a loss profile calibrated to the timing and severity of consumer-loan defaults they are profitable. What survives every specification is the *ordering*: within the passive regimes, out-of-sample AUC and profit are unrelated (the highest-AUC scorecard is the least profitable), and an active mechanism whose own ranking score has the lowest AUC in the study creates 1.4 to 1.5 times the value of the best passive regime. The paradox is therefore not that machine learning loses money; it is that a classifier used only as a filter leaves roughly a third of the attainable value unexploited, because it has no instrument for pricing, re-structuring, or capital allocation.

The paradox has a simple formal core. Let p_i be the PD that a model assigns to borrower i. A passive regime prices nothing: it approves the original contract if and only if its expected value covers the acquisition cost, and its profit is the sum over the approved borrowers,

$$a_i(p) = \mathbf{1}\{NPV_i(p_i) > CAC_i\}, \qquad NPV_i(p) = -L_i + (1-p)\,PV_{nd,i} + p\,PV_{d,i},$$
$$\Pi(p) = \textstyle\sum_i a_i(p)\,[NPV_i(p_i) - CAC_i].$$

Because default destroys value ($PV_{nd,i} > PV_{d,i}$), NPV_i is decreasing in p, and every approval compares the *level* p_i with a borrower-specific breakeven PD p_i^* that solves $NPV_i(p_i^*) = CAC_i$. The AUC, by contrast, is the probability that a randomly drawn defaulter is scored above a randomly drawn non-defaulter, so it depends on the scores only through their order. For every strictly increasing transformation g of the scores,

$$\text{AUC}(p) = \Pr\left(p_i > p_j \mid D_i = 1,\ D_j = 0\right),$$
$$\text{AUC}(g \circ p) = \text{AUC}(p), \qquad \text{but in general} \qquad \Pi(g \circ p) \neq \Pi(p).$$

Remark (criterion mismatch). The AUC depends on the scores alone: it involves neither the principal L_i, nor the rate R_i, nor the maturity, nor the acquisition cost CAC_i of any loan, all of which enter Π. Because every strictly increasing transformation of an AUC-maximizing score is again an AUC maximizer, the set of

AUC maximizers contains scores that earn different profits, so that in general

$$\arg\max_{p} \ \mathrm{AUC}(p) \ \not\subseteq \ \arg\max_{p} \ \Pi(p).$$

Two PD models with the same AUC can therefore earn different profits, and a model with a higher AUC can earn less: discrimination is neither necessary nor sufficient for value. The active engine adds a second, separate source of the gap. Its choice set contains the original contract, which is accepted with certainty, so in its own objective (capital-adjusted value net of acquisition cost) it weakly dominates, borrower by borrower, the passive rule built on the same PD, and it gains strictly whenever re-pricing or re-structuring a contract is worth more than accepting or rejecting it as given (Section 4.7).

The data contain a natural experiment for the first statement. Platt scaling (Section 3.3) is a strictly increasing map of the Random Forest's raw scores, so the raw and the calibrated forest have the same AUC, 0.650, by construction. The passive +NAM regime nevertheless earns \$6.61M with the raw forest and \$8.51M with the calibrated one, and the engine \$10.38M and \$12.55M (Sections 5.3 and 5.10). The ranking is identical; only the probability levels differ, and so does the profit.

1.3 Central Research Question

> **How can an active, multi-stage mechanism co-optimize contract terms (amount, rate, maturity), customer acquisition cost, and behavioral acceptance so as to maximize a lender's Basel III/IV risk-adjusted return on capital, when borrower default responds endogenously to the prices the mechanism sets, when acceptance responds to those same prices, and when every cash flow is valued at present value under a realistic loss profile?**

The two endogeneity clauses are the theoretical spine of the thesis. If default were exogenous to price, the optimal policy would hike every rate to the demand ceiling; if acceptance were exogenous, it would hike further still. Stiglitz and Weiss (1981) show that the first is false: raising the rate adversely selects the pool and strains affordability, so PD is increasing in the very lever the optimizer wishes to pull. Demand theory shows the second is false: the borrower who is offered a worse contract may walk. The engine is built to respect both, and Section 5 shows how much of its advantage each one removes.

The question is operationalized as four hypotheses, each tested against a pre-specified base case and a battery of robustness checks.

- **H1 (value).** The engine's expected net profit exceeds that of every passive regime, including passive regimes given the same acquisition-margin and capital screens; the difference is statistically distinguishable from zero and survives re-estimation on independent partitions of the data (Sections 5.1, 5.2, 5.11).
- **H2 (discrimination is not value).** Among passive regimes, out-of-sample discriminatory power (AUC) does not order profitability, and the AUC differences between the passive scorecards are small relative to their sampling error (Section 5.3).

- **H3 (mechanism).** The engine's advantage is generated by price-setting rather than by changes in principal or maturity, and it is bounded by the demand elasticity in the way Proposition 1 predicts (Sections 5.4, 5.5, 5.9).
- **H4 (robustness).** The ranking of the engine over the best passive regime is invariant to the loss profile, the asset correlation, the PD model, the acquisition-cost assumption and the identification of the rate channel (Sections 5.7, 5.6, 5.10).

1.4 Contributions

The thesis makes five contributions.

1. **Empirical endogenization of default through Stiglitz-Weiss channels.** The interest-rate and affordability elasticities of default are estimated from loan-level data and embedded in a dynamic logit that responds to every candidate contract's rate and payment burden; a within-sub-grade (fixed-effects) estimate is used to bound the identification of the rate channel.
2. **A capital-aware contract optimizer with a realistic loss profile, and its analytical characterization.** The optimizer values 1,178 candidate contracts per borrower under discounted French amortization with an explicit time-to-default and loss-given-default, and charges each candidate a Basel ASRF capital cost at a hurdle rate, so that pricing and capital allocation are decided jointly rather than sequentially. An inverse semi-elasticity pricing rule (Proposition 1) characterizes the optimum and explains when the engine re-prices, when it does not, and why its hikes cluster at the regulatory-style cap.
3. **A screening mechanism with heterogeneous, partially observed demand.** Offers are routed through a three-tier bargaining tree in which the lender's belief about the borrower's price sensitivity is updated after a rejection, in the tradition of second-degree price discrimination (Mussa & Rosen, 1978; Rochet & Choné, 1998).
4. **Pre-sourcing Net Acquisition Margin gating on mechanism value.** Marketing spend is committed only when the expected value of the whole bargaining tree, not merely of the most aggressive offer, covers the borrower's acquisition cost.
5. **A profit-based, capital-based and inference-based evaluation.** Ten regimes are evaluated out of sample under identical accept rules with expected, pseudo-realized and realized profit and Vasicek economic capital; the engine's advantage is given a paired-bootstrap confidence interval, re-estimated on twenty independent partitions, decomposed by contract axis and by model feature, and examined for who bears the re-pricing.

---

2. Theoretical Framework and Literature Review

2.1 Credit Rationing and Endogenous Risk Pricing (Stiglitz & Weiss, 1981)

In credit markets with asymmetric information the interest rate is not a market-clearing price and cannot be treated as a free profit lever. Stiglitz and Weiss (1981) identify two channels through which a higher contractual rate raises the lender's realized default rate, so that expected return is non-monotonic in price. Through *adverse selection* the rate acts as a screening device: as it rises, safe types who know their own risk and have alternatives leave the pool, while risky types, less sensitive to price because they anticipate not repaying, remain. Through *moral hazard and affordability* a higher required payment raises the debt-service burden and, in consumer credit, mechanically raises the probability that the household cannot service the loan. The expected-return-maximizing rate is therefore interior, and lenders may ration credit rather than clear the market with price. The engine operationalizes both channels by making PD an explicit, estimated, increasing function of the rate shock and of the payment-to-income shock (Section 3.2), which is why the naive rate-hiking strategy is suboptimal in this framework and why the demand side (Section 4.5) is the natural counterweight.

Empirically separating the two channels is difficult because, in observational data, lenders set rates as a function of perceived risk. Karlan and Zinman (2009) solve the problem with a field experiment that randomizes the offered and the contracted rate separately, and so distinguishes selection on the offer from the effect of the repayment burden; Adams, Einav and Levin (2009) exploit pricing and down-payment variation in subprime auto lending; and Dobbie and Skiba (2013) use discontinuities in payday-loan eligibility to show that a larger loan, holding the borrower fixed, *lowers* default, while borrowers who *choose* larger loans are adversely selected, a clean demonstration that the causal and the selection effects of contract terms can have opposite signs. These designs are not available in our data, which is why Section 3.2 treats the estimated rate elasticity as an upper bound and Section 5.10 re-estimates it within sub-grades.

2.2 Structural Portfolio Credit Risk and Regulatory Capital (Merton 1974; Vasicek 2002; Basel ASRF)

Merton (1974) prices default as the event that a latent asset value falls below a liability threshold. Vasicek (2002) extends this to a large homogeneous portfolio in which every obligor's asset return loads on one common factor $Z \sim N(0,1)$ with asset correlation ρ, yielding the *Asymptotic Single-Risk-Factor* (ASRF) loss distribution that underlies the Basel II/III Internal Ratings-Based risk-weight functions (BCBS, 2005; Gordy, 2003). The economically crucial object is the conditional default probability

$$PD^{cond}(Z) = \Phi\left(\frac{\Phi^{-1}(PD) - \sqrt{\rho}\,Z}{\sqrt{1-\rho}}\right),$$

whose 99.9% quantile defines the capital requirement $K(PD) = LGD\,[PD^{cond}(Z_{0.999}) - PD]$. Two aspects of the Basel treatment are used here that the proposal did not use. First, the regulatory asset correlation for unsecured retail ("other retail") is not a constant but a decreasing function of PD, $\rho(PD) = 0.03\,w + 0.16\,(1-w)$ with $w = (1-e^{-35PD})/(1-e^{-35})$ (BCBS, 2005; Basel Committee on Banking Supervision,

2019, CRE31); the constant $\rho = 0.15$ of the proposal is the residential-mortgage value and is retained only as a stress case. Second, the capital requirement is not merely a reporting statistic: a lender that must hold K against an exposure incurs a cost equal to the hurdle return on that capital, and a contract optimizer that ignores this cost will systematically prefer high-PD, high-rate contracts whose capital consumption is not paid for. Section 4.6 puts the charge inside the objective.

2.3 Mechanism Design and Screening under Demand Uncertainty (Mussa & Rosen 1978; Rochet & Choné 1998)

The engine's sequential bargaining is, formally, a screening problem in the tradition of second-degree price discrimination. Mussa and Rosen (1978) characterize a monopolist offering a menu of quality-differentiated products to privately informed consumers; Rochet and Choné (1998) extend screening to multidimensional types. In our setting the product is a triple (principal, maturity, rate), the type is the borrower's private price sensitivity, and the lender designs offers subject to *individual rationality* (the borrower can always retain the original contract, so no offer can make the borrower worse off) and *incentive compatibility* (the borrower who declines the aggressive contract is offered the contract that maximizes expected surplus over the fallback). The version implemented here adds one feature that the proposal lacked: the lender does not know the borrower's type but holds a prior over two types, and a rejection is informative. The Tier-2 offer is therefore chosen under the *posterior* belief, and the contract that was just rejected is never re-offered. This is the simplest form of Bayesian screening and is what turns the three tiers from a fixed script into a mechanism.

2.4 Non-Linear Endogenous Customer Acquisition Cost

Standard capital budgeting records customer acquisition as a fixed overhead. For a digital lender whose marketing spend is auction-priced and convex in the target's attractiveness this is inadequate. We model CAC as an endogenous, non-linear function of the contract (Section 4.1): larger tickets require disproportionately more marketing because the pool of qualified high-balance borrowers is thinner, and sourcing borrowers into higher-priced offers raises drop-off and hence cost per funded loan. Treating CAC endogenously couples the sourcing decision to pricing, which is what permits the pre-sourcing gate to *avoid* spend on structurally value-negative prospects rather than sink it.

2.5 Profit-Based Evaluation of Credit Models (Hand 2009; Verbraken et al. 2014; Serrano-Cinca & Gutiérrez-Nieto 2016)

Three strands of the evaluation literature frame Section 5. Hand (2009) shows that the AUC is an incoherent measure of classifier performance for decision-making because it implicitly averages over a distribution of misclassification costs that depends on the classifier itself; a lender faces one cost structure, not all of them. Verbraken et al. (2014) derive an expected-maximum-profit measure for credit scoring and show that model rankings under it differ from AUC rankings. Serrano-Cinca and Gutiérrez-Nieto (2016) apply profit scoring directly to peer-to-peer lending data (the same LendingClub environment as ours) and find that the variables

that predict *profit* are not the variables that predict *default*. Our contribution to this strand is to take the profit measure one step further: not merely to evaluate a fixed contract by profit, but to let the profit objective choose the contract. Calibration, as distinct from discrimination, then becomes essential (Platt, 1999; Niculescu-Mizil & Caruana, 2005), because a contract optimizer consumes the *level* of PD, not only its ranking.

The machine-learning literature on credit scoring supplies the other side of the comparison. Khandani, Kim and Lo (2010) show that machine-learning forecasts of consumer delinquency add value over traditional scores; Lessmann, Baesens, Seow and Thomas (2015) benchmark forty-one classifiers on eight credit data sets and find that ensembles such as random forests are strong but that gains over logistic regression are often modest; Jagtiani and Lemieux (2019) document on LendingClub data that the platform's own grades embed information beyond FICO; and Berg, Burg, Gombović and Puri (2020) show that even digital-footprint variables rival credit-bureau scores. Fuster, Goldsmith-Pinkham, Ramadorai and Walther (2022) add the distributional caveat that better statistical technology changes *who* gains and loses from credit pricing. Our results sit at the intersection: the Random Forest here is a competent but not dominant scorer, exactly as Lessmann et al. would predict, and what matters for value is how its output is used.

2.6 Contract Pricing and Demand in Consumer Credit (Einav, Jenkins & Levin 2012; Attanasio, Goldberg & Kyriazidou 2008)

A contract optimizer is only as credible as its demand side. Einav, Jenkins and Levin (2012) estimate the joint effect of price and down payment on take-up and default in subprime auto credit and show that contract terms other than the interest rate are powerful screening and pricing instruments; Attanasio, Goldberg and Kyriazidou (2008) find that, except for high-income households, consumers are very responsive to loan maturity and less responsive to the interest rate, the signature of binding liquidity constraints. Phillips (2005) sets out the revenue-management approach to consumer-lending prices, in which acceptance probabilities are estimated from offer-and-response data and the price maximizes expected margin times take-up, the structure of Section 4.5. Two consequences follow for this thesis. First, our acceptance function carries no explicit preference for longer maturity, so the finding that the engine never shortens a loan (Section 5.4) is a statement about the default channel, not about borrowers' maturity preferences; if borrowers are as maturity-sensitive as Attanasio et al. find, shortening would also be penalized on the demand side and the finding would only be reinforced. The grid offers no maturity longer than 60 months, the direction in which that literature suggests value might lie. Second, because the demand parameters cannot be estimated without offer-and-response data, they are reported as a breakeven analysis rather than as point estimates (Section 5.9).

---

3. Data and Empirical Estimation

3.1 Dataset, Descriptive Statistics, and Partitioning

The empirical analysis uses, as its testbed, a LendingClub-style consumer-loan population of **10,000 loans** with loan-level covariates: principal (`loan_amnt`), contractual rate (`int_rate`), credit grade and sub-grade, employment length, revolving balance and utilization, FICO score, home ownership, annual income, verification status, and a binary default outcome. All loans are placed on a common 60-month horizon for cash-flow comparability. The overall default rate is 16.23%. Marketplace consumer loans are a convenient testbed for three reasons: the data are public and widely used in the credit-scoring literature (Serrano-Cinca & Gutiérrez-Nieto, 2016; Jagtiani & Lemieux, 2019; Croux et al., 2020), so results can be compared and replicated; the population is unsecured, fixed-rate and fully amortizing, the product for which contract re-pricing is most natural; and the platform's own grades supply a realistic incumbent scorecard against which to benchmark. Nothing in the engine is specific to this population.

Table 0. Sample descriptive statistics (full population, $N = 10{,}000$).

Variable	Mean	Std. Dev.	Median	Min	Max
Loan amount ($)	12,221.87	7,196.99	10,000	1,000	35,000
Interest rate (%)	13.20	3.93	13.11	6.03	25.80
FICO score	698.88	30.60	692.0	662.0	847.5
Annual income ($)	68,979.54	47,880.29	60,000	7,000	1,200,000
DTI proxy (%)	23.47	18.06	19.79	0.00	436.68
Payment-to-income	0.05	0.03	0.05	0.00	0.16
Default indicator	0.1623	0.369	0	0	1

Population anchors used by the acquisition-cost function: $\bar{L} = \$12{,}221.87$, $\bar{R} = 13.20\%$.

Two data-preparation steps precede estimation. Missing employment length is coded as an explicit "Unknown" level and missing revolving utilization is imputed at the median. Five sub-grades with fewer than ten loans in the population (F4, F5, G1, G2, G5; ten loans in total) are pooled into a single level, and the two rare home-ownership levels (NONE, OTHER) are merged, so that every factor level present in any test partition is also present in the corresponding training partition. This pooling is required for the repeated-split analysis of Section 5.11 and shifts the base-case numbers by about one percent relative to an unpooled run.

The population is partitioned into an 80% training set (8,000 loans) and a 20% test set (2,000 loans) under a fixed seed. Every quantity that is estimated, the Random Forest, its calibration map, the grade, sub-grade, FICO and integrated logistic scorecards, and the Stiglitz-Weiss default regression, is estimated on the training set only and evaluated on the held-out test set. Portfolio aggregates are scaled by five to express a 10,000-loan book. Section 5.11 repeats the entire pipeline on twenty alternative partitions to show that no conclusion depends on the canonical seed.

3.2 The Empirical Stiglitz-Weiss Default Model (`fit_emp`)

To endogenize default we estimate, on the training set, a logistic regression of the default indicator on the two Stiglitz-Weiss channels and three credit controls:

$$\ln\left(\frac{PD_i}{1-PD_i}\right) = \beta_0 + \gamma_{emp}\,\text{int_rate}_i + \delta_{emp}\,\text{pmt_to_inc}_i + \beta_3\,\text{fico}_i + \beta_4\,\text{dti}_i + \beta_5\,\text{revol_util}_i,$$

where $\text{pmt_to_inc}_i = PMT_i^{orig}/(Y_i/12)$ is the original monthly Spitzer payment scaled by monthly income and $\text{dti}_i = (\text{revol_bal}_i/Y_i) \times 100$ is a revolving-debt-to-income proxy (the source data contain no native DTI field).

Table 1. Empirical Stiglitz-Weiss default regression (`fit_emp`), estimated on 8,000 training loans.

Regressor	Estimate	Std. Error	z	Pr($>\|z\|$)	
(Intercept)	-0.175798	1.232645	-0.143	0.88659	
int_rate (γ_{emp})	$+$**0.089788**	0.010440	8.601	7.92×10^{-18}	***
pmt_to_inc (δ_{emp})	$+$**6.574745**	1.092396	6.019	1.76×10^{-9}	***
fico	-0.004403	0.001609	-2.737	0.00621	**
dti	$+0.005359$	0.001678	3.193	0.00141	**
revol_util	-0.002738	0.001641	-1.669	0.09521	·

Null deviance $= 7054.6$ *(7999 df); residual deviance* $= 6781.8$ *(7994 df); AIC* $= 6793.8$. *Significance:* $^{***}p < 0.001$, $^{**}p < 0.01$, $^{\cdot}p < 0.1$.

Each additional percentage point of contractual rate raises the log-odds of default by 0.090, an odds multiplier of $e^{0.0898} \approx 1.094$; a five-percentage-point rise in payment-to-income multiplies the odds by $e^{6.575 \times 0.05} \approx 1.39$. The controls carry their theoretical signs.

> **Identification caveat.** The rate coefficient should be read as *consistent with*, not proof of, the Stiglitz-Weiss adverse-selection channel. Contractual rates are risk-based-priced, so `int_rate` and unobserved risk are jointly determined; without an instrument or a pricing discontinuity (Adams, Einav & Levin, 2009) the coefficient blends a behavioral response with a risk-based-pricing correlation and is best read as an upper bound on the pure behavioral channel. For the engine this is conservative in the right direction: it makes the optimizer more, not less, cautious about uncompensated rate hikes, and the tornado analysis of Section 5.8 shows that a 25% smaller γ_{emp} would *raise* the engine's profit.

3.3 Probability-of-Default Models and Calibration

Four passive PD models are estimated on the training set: a FICO-only logit, a grade logit, a sub-grade logit, and a Random Forest (500 trees, eleven covariates). An integrated logistic rating on the same eleven covariates is estimated as a discrimination ceiling but is not used as a lending regime. Because the engine consumes the *level* of PD, the Random Forest's raw vote share is calibrated by Platt scaling (Platt, 1999): a logistic map $\sigma(a + b\,\mathrm{logit}(\hat{p}))$ is fitted to *out-of-fold* Random Forest predictions on the training set (five folds), so that the calibration map never sees a prediction made on data the forest was trained on. The fitted map has intercept -0.797 and slope 0.559, confirming that raw Random Forest vote shares are over-dispersed (too extreme in both tails), as Niculescu-Mizil and Caruana (2005) document for bagged trees.

Table 1b. Out-of-sample discrimination and calibration of the PD models (test set, $n = 2{,}000$, observed default rate 16.85%).

Model	Brier	AUC	Calibration intercept	Calibration slope	Mean predicted PD
Random Forest, raw	0.1361	0.6497	−0.524	0.705	0.1856
Random Forest, Platt (used by the engine)	0.1351	0.6497	0.481	1.261	0.1596
Integrated logistic rating	0.1335	0.6691	0.210	1.105	0.1620
FICO-only	0.1366	0.6285	0.848	1.499	0.1613
Grade	0.1354	0.6419	0.392	1.225	0.1630
Sub-grade	0.1343	0.6597	0.454	1.266	0.1627

Calibration intercept and slope are from a logistic regression of the test outcome on the logit of the predicted PD; perfect calibration is $(0, 1)$. *Platt scaling leaves the AUC unchanged (it is monotone) and improves the Brier score and the mean level. On this test set the map somewhat over-corrects (slope 1.26 rather than 1.00): a limitation noted in Section 7.2, and one that biases the engine toward caution, since it compresses the PD differences the optimizer acts on.*

---

4. Methodology and Algorithmic Pipeline

The complete specification follows. Base-case parameters are listed in Appendix B. Relative to the proposal (and to the frozen prototype `sidur_12.R`, whose results are reproduced in Section 4.11), seven features are new; each is marked **[F1]**–**[F7]** where it enters. The pipeline is modular: the PD model of Section 3.3 is a plug-in, and Section 5.10 re-runs the engine with two alternative PD models to show that the results do not depend on the choice.

4.1 Non-Linear Endogenous CAC

$$CAC_i = \kappa\, L_i \left(\frac{L_i}{\bar{L}}\right)^{\phi} \left(\frac{R_i}{\bar{R}}\right)^{\gamma_{cac}},$$

with $\kappa = 0.02$ the base marketing rate, $\phi = 0.20$ the loan-size friction elasticity and $\gamma_{cac} = 0.15$ the rate-aversion elasticity; L_i, R_i are the borrower's original principal and rate. CAC is a function of the *original* contract because it is incurred before the offer is made.

4.2 Discounted Spitzer Amortization with an Explicit Loss Profile [F1]

With monthly contract rate $r_c = R/1200$, monthly cost of funds $r_m = r_{disc}/12$ ($r_{disc} = 5\%$) and term n, the constant payment and the outstanding balance after t payments are

$$PMT = \frac{L\, r_c}{1-(1+r_c)^{-n}}, \qquad B_t = L(1+r_c)^t - PMT\,\frac{(1+r_c)^t - 1}{r_c}.$$

The proposal valued a defaulted loan as an immediate total loss of principal. Consumer-loan defaults do not occur at origination and do not lose everything: a defaulting borrower typically makes a number of payments first, and part of the outstanding balance is recovered. We therefore introduce two parameters, a time to default $\bar{\tau}$ (months; base 12, capped at n) and a loss given default LGD (base 0.8) applied to the balance outstanding when payments stop. The present value of the contract to the lender conditional on no default and conditional on default are

$$PV_{nd} = PMT\, a(n, r_m), \qquad a(k,r) = \frac{1-(1+r)^{-k}}{r},$$
$$PV_d = PMT\, a(\bar{\tau}-1, r_m) + (1-LGD)\, B_{\bar{\tau}-1}\,(1+r_m)^{-\bar{\tau}},$$

and the expected and realized net present values are

$$NPV_{cond} = -L + (1-PD)\, PV_{nd} + PD\, PV_d, \qquad NPV_{real} = -L + (1-D)\, PV_{nd} + D\, PV_d,$$

with D the observed default indicator. The exposure at default used by the loss simulation is $EAD = LGD \cdot B_{\bar{\tau}-1}$. The proposal's specification is the special case $\bar{\tau} = 1$, $LGD = 1$, which is retained throughout as the *conservative bound* (Section 5.7). The base values are chosen conservatively against the evidence available for unsecured consumer instalment credit. On recoveries, Zhang and Thomas (2012) report a mean recovery rate of 0.42 (an LGD of about 0.58) on 27,278 defaulted UK unsecured personal loans; recoveries on marketplace loans are lower, because charged-off LendingClub loans are sold or collected against fees of up to 35% of amounts recovered (LendingClub Corporation, 2017; see also Chen, Lo & Tang, 2022, who assume total loss).

An LGD of 0.8 therefore sits between the bank benchmark and the zero-recovery bound. On timing, we found no verifiable public statistic on months-to-default for this population; longer terms are known to default more (Croux, Jagtiani, Korivi & Vulanovic, 2020), and $\bar{\tau}$ is best regarded as a scenario parameter. Because these two assumptions are the most influential in the model, Section 5.7 reports a six-point grid over them, from the immediate-total-loss bound ($\bar{\tau} = 1$, $LGD = 1$) to a lenient profile ($\bar{\tau} = 24$, $LGD = 0.6$).

4.3 A Geometric Interpretation of the Contract

The three quantities the engine manipulates, principal, maturity and rate, admit a representation that clarifies what the optimizer is doing. Consider the balance schedule B_t of a contract (L, R, T) as the profile of a solid whose height at month t is the outstanding balance and whose depth along a third axis is the contractual rate R (Figure 9a). Under French amortization the profile is concave, falling slowly at first and steeply at the end, so the solid is a blunt "bullet", not a cone: at month 12 of a 60-month loan the median borrower still owes 86% of the principal. Three measures of this solid are exactly the three value components in NPV_{cond}. Its *volume*, $V = \sum_{t<T} B_t$, is the loan's exposure in dollar-months; the interest income is $(R/12)\,V$, so the rate axis scales the volume into revenue. Its *discounted volume per dollar*, $X = \frac{1}{12}\sum_{t<T}(B_t/L)\,v^t$ with $v = (1 + r_m)^{-1}$, is the exposure-weighted duration that multiplies the capital requirement in Section 4.6. And a default at $\bar{\tau}$ is a *guillotine* that truncates the solid: the slice removed is the exposure at default $LGD \cdot B_{\bar{\tau}-1}$, and each monthly repayment before it is a thin slice cut from the base (Figure 9b).

In this language the engine does not choose a point but a *shape*. Cutting the principal shrinks the base; shortening the term lowers the height and steepens the profile (raising every slice and therefore the payment-to-income ratio); raising the rate deepens the solid and, through the Stiglitz-Weiss response, brings the guillotine's probability forward (Figure 9c). The distribution of chosen shapes in Section 5.4 is then easy to read: the engine leaves the base and the height alone and works almost entirely on the depth. Exogenous shocks enter the same picture in two distinct ways (Figure 9d). A change in the lender's cost of funds does not change the solid, whose cash flows are contractually fixed, but re-weights its slices through v^t; the tornado analysis shows this is the second-largest sensitivity in the model. A macroeconomic shock enters through the Vasicek factor Z, which moves the position of every guillotine in the book at once; that is what the economic-capital simulation of Section 4.10 measures. What the picture does *not* contain, prepayment and refinancing, which would remove slices from the top of the solid, is noted as an extension in Section 7.3.

Figure 9. Geometric interpretation of a loan contract and of the engine's search (illustrative median borrower)

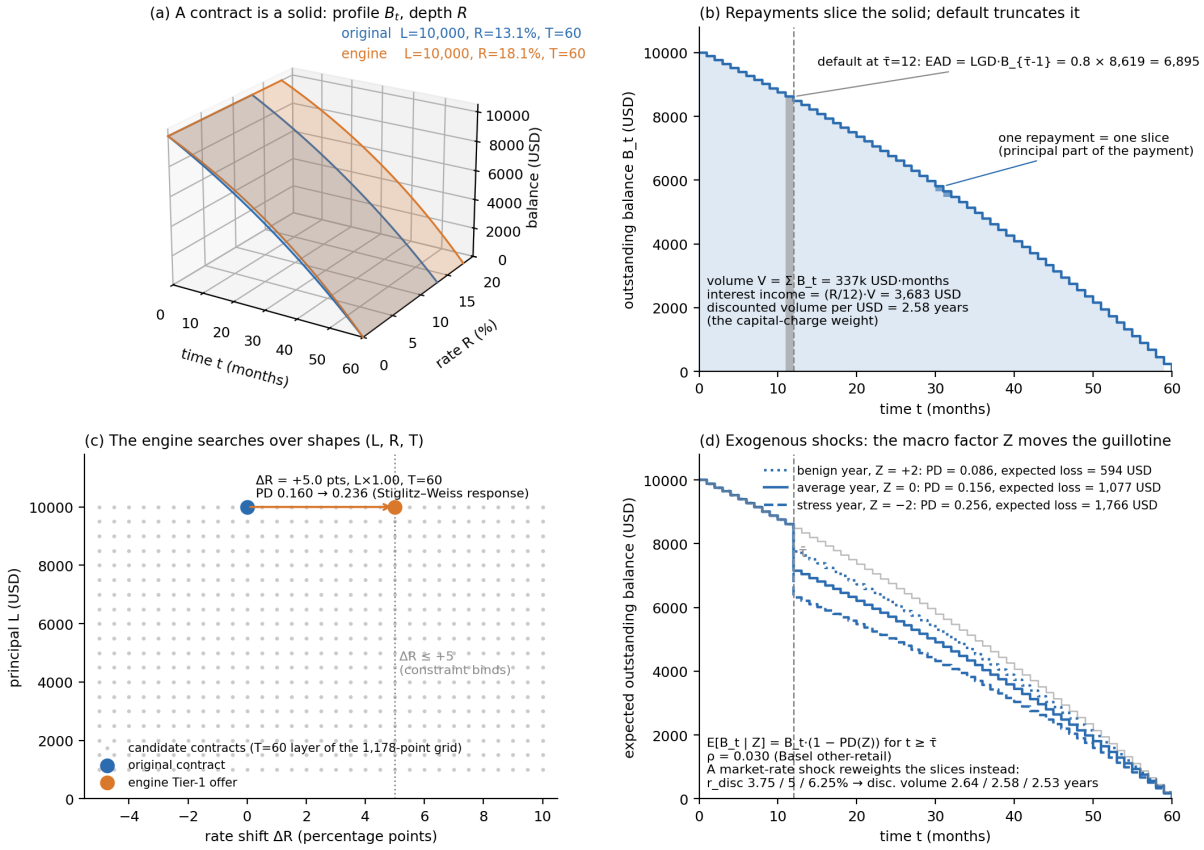


Figure 9. Geometric interpretation of a loan contract and of the engine's search (illustrative median borrower: \$10,000 at 13.1%, 60 months, PD 0.16).

4.4 Endogenous Dynamic Logit Default

For a candidate contract (L_c, R_c, T_c) with payment PMT_c on a borrower with calibrated baseline PD PD_{base} and monthly income $Y/12$,

$$\text{logit}(PD_{dyn}) = \text{logit}(PD_{base}) + \gamma_{emp}\,(R_c - R_{orig}) + \delta_{emp}\,\frac{PMT_c - PMT_{orig}}{Y/12},$$
$$PD_{dyn} \in (10^{-4}, 1 - 10^{-4}).$$

where $\text{logit}(p) = \ln[p/(1-p)]$ and its inverse is $\text{logit}^{-1}(x) = 1/(1+e^{-x})$, so that $PD_{dyn} = \text{logit}^{-1}(\cdot)$ with the clipping shown. Principal cuts and term changes enter through PMT_c, so the affordability channel subsumes leverage and duration effects; because the level of `int_rate` is already in PD_{base}, only the shock enters, avoiding double counting.

4.5 Customer Acceptance with Heterogeneous Elasticity and Two Borrower Types [F3, F6]

The proposal used a single hard-coded hike-aversion constant. Two changes make the demand side explicit. First, hike aversion is a named parameter η_0 (base 0.05 per percentage point) and varies across borrowers with their credit quality,

$$\eta_i = \eta_0 \min\left(\eta_{cap}, \left(\frac{PD_{ref}}{PD_{base,i}}\right)^{\xi}\right), \qquad \xi = 0.5,\ \eta_{cap} = 4,$$

with PD_{ref} the test-set median PD: a borrower with a quarter of the median PD is twice as price-sensitive, a borrower with four times the median PD half as sensitive. This is the demand-side face of adverse selection: safe borrowers have outside options and walk from a hike, risky borrowers do not. Second, within a credit quality, borrowers are of two unobserved types, $k \in \{\text{low}, \text{high}\}$ with sensitivity multipliers $m_k \in \{0.5, 1.5\}$ and prior $\pi_k = (0.5, 0.5)$, chosen mean-preserving so that the average elasticity is η_i. The type-k acceptance probability of a contract is

$$p_k = \text{clip}_{[0,1]}\left[\left(\frac{L_c}{L_{orig}}\right)^{\alpha} e^{-\lambda \max(0, PMT_c - PMT_{orig})}\, \rho_k\right],$$

$$\rho_k = \begin{cases} e^{-\eta_i m_k (R_c - R_{orig})}, & R_c > R_{orig} \\ 1 + \eta_{cut}(R_{orig} - R_c), & R_c \le R_{orig}, \end{cases}$$

with $\alpha = 1$, $\lambda = 0.003$ and $\eta_{cut} = 0.02$; the prior-weighted acceptance is $\bar{p} = \sum_k \pi_k p_k$. The demand side remains calibrated rather than estimated, so Section 5.9 reports the *breakeven* elasticity, the value of η_0 at which the engine's advantage would vanish.

4.6 Capital-Aware Objective and Deployability Constraints [F5, F7]

A contract that earns a positive expected NPV but consumes capital on which the lender must earn a hurdle return h is not necessarily worth writing. Following the Basel ASRF treatment, every candidate contract is charged

$$C^{cap} = h \cdot K(PD_{dyn}) \cdot L_c \cdot X(L_c, R_c, T_c),$$

$$K(PD) = LGD\left[\Phi\left(\frac{\Phi^{-1}(PD) + \sqrt{\rho(PD)}\,\Phi^{-1}(0.999)}{\sqrt{1-\rho(PD)}}\right) - PD\right],$$

with $h = 10\%$, $\rho(PD)$ the Basel other-retail correlation of Section 2.2, and X the discounted exposure duration of Section 4.3, so that the charge is the hurdle return on the average capital held over the life of the contract. The objective the optimizer maximizes is the acceptance-weighted *value* $\bar{p}\,(NPV_{cond} - C^{cap})$. Because K is convex in PD, the charge penalizes exactly the contracts a naive NPV maximizer likes best, high-

rate offers to high-PD borrowers whose systematic risk is not paid for; Section 5.6 shows it lowers the engine's approval rate by six points and raises its RAROC.

Three constraints restrict the feasible set to contracts a consumer lender could actually offer: the rate hike is capped at $\Delta R \leq +5$ percentage points; the new rate may not exceed 1.5 times the original; and the new payment-to-income ratio may not exceed $\max(0.15,\ PTI_{orig})$, so that no offer pushes a borrower above a 15% payment burden unless they were already there. Section 5.1 reports the unconstrained engine alongside the constrained one; the constraints cost about 7% of expected profit and are the price of deployability.

4.7 Grid Search, NAM Gate and the Three-Tier Bargaining Tree [F2, F4, F6]

For each borrower the optimizer evaluates the full grid, principal fraction $\in \{1.00, 0.95, \ldots, 0.10\}$ (19 values) $\times$ maturity $\in \{36, 60\}$ $\times$ rate shift $\in \{-5.0, \ldots, +10.0\}$ in 0.5-point steps (31 values), **1,178 candidate contracts**, and tracks two of them. The *aggressive* contract maximizes the acceptance-weighted value over the feasible set,

$$\text{best} = \arg\max_{\text{grid}}\ \bar{p}\,(NPV_{cond} - C^{cap}),$$

and the *counter-offer* maximizes expected surplus over the borrower's fallback, $S = p_2\,[(NPV_{cond} - C^{cap}) - (NPV_{orig} - C^{cap}_{orig})]$ subject to $S > 0$, where p_2 is the acceptance probability under the lender's belief at the time the counter-offer is made. If the aggressive contract was offered and declined, that belief is the Bayesian posterior over types given the rejection,

$$\pi_k^{post} = \frac{\pi_k\,(1 - p_k^{best})}{\sum_j \pi_j\,(1 - p_j^{best})},$$

which shifts weight toward the price-sensitive type and therefore toward gentler counter-offers; the rejected contract itself is excluded from the Tier-2 search. Borrowers are then routed by the sign of their original contract's NPV.

Group 1 ($NPV_{orig} > 0$). Let $a = \mathbb{1}[\bar{p}^{best}(NPV^{best} - C^{cap,best}) > NPV_{orig} - C^{cap}_{orig}]$ indicate that the aggressive contract is worth offering. Tier 1 offers it; if it is declined, a fraction ω of borrowers leave (attrition; base $\omega = 0$, stress-tested up to 0.5), the remainder receive the counter-offer, and Tier 3 retains the original contract. The route probabilities are

$$P_1 = a\,\bar{p}^{best}, \qquad P_w = a\,(1 - \bar{p}^{best})\,\omega,$$
$$P_2 = (1 - P_1 - P_w)\,p_2^{indiff}\,\mathbb{1}[S^{indiff} > 0], \qquad P_3 = 1 - P_1 - P_2 - P_w,$$

and the borrower's expected profit and *mechanism value* are

$$\begin{aligned}\mathbb{E}[\pi_i] &= P_1\, NPV_{cond}^{best} + P_2\, NPV_{cond}^{indiff} + P_3\, NPV_{orig},\\ \mathcal{V}_i &= P_1(NPV_{cond}^{best} - C^{cap,best}) + P_2(NPV_{cond}^{indiff} - C^{cap,indiff})\\ &\quad + P_3(NPV_{orig} - C_{orig}^{cap}).\end{aligned}$$

Group 2 ($NPV_{orig} \le 0$). The borrower is offered the engineered contract only and is rescued with probability $\bar{p}^{best}$.

The NAM gate. Marketing spend is committed only if the value the lender expects from the whole tree exceeds the acquisition cost: for Group 1, $\mathcal{V}_i \ge CAC_i$; for Group 2, $\bar{p}^{best}(NPV^{best} - C^{cap,best}) \ge CAC_i$. The proposal gated on the aggressive contract alone, which admitted borrowers whose realistic expected value (mostly Tier 3) could not cover their CAC. In the base case the gate sources 1,732 of 2,000 test borrowers (1,710 in Group 1 and 22 in Group 2, of whom 16.4 are expected to be rescued); the remaining 268 are never marketed to and their CAC is avoided. Algorithm 1 summarizes the per-borrower procedure.

```
Algorithm 1. The multi-stage engine for one borrower i
Input: original contract (L0, R0, T0 = 60), income Y, calibrated
       PD_base, CAC_i, estimated (gamma, delta), demand and capital
       parameters
 1 for each candidate c in the 1,178-point grid:
 2     PMT(c), B_t(c)  <- French amortization          (Sec. 4.2)
 3     PD_dyn(c)       <- Stiglitz-Weiss logit shock   (Sec. 4.4)
 4     NPV_cond(c)     <- loss profile                 (Sec. 4.2)
 5     C_cap(c)        <- Basel capital charge         (Sec. 4.6)
 6     p_k(c), k = low/high;  pbar(c) = sum_k pi_k p_k(c)  (4.5)
 7     feasible(c)     <- dR <= 5, R <= 1.5 R0,
                          PTI <= max(0.15, PTI0)       (Sec. 4.6)
 8 best   <- argmax over feasible c of
             pbar(c) * [NPV_cond(c) - C_cap(c)]        (Tier 1)
 9 post   <- Bayes update of pi_k after a rejection of best
10 indiff <- argmax over feasible c != best of
             p2_post(c) * [value(c) - value(orig)]     (Tier 2)
11 compute P1, Pw, P2, P3 and mechanism value V_i      (Sec. 4.7)
12 if NPV_orig > 0: source i iff V_i >= CAC_i          (Group 1)
   else:            source i iff pbar(best) * value(best) >= CAC_i
Output: offers (best, indiff, orig), route probabilities,
        E[pi_i], and the PD and EAD of each route
```

4.8 An Analytical Pricing Rule

The grid search finds the optimum numerically, but its structure can be characterized analytically, which both explains the engine's behavior (Section 5.4) and disciplines the demand-calibration debate (Section 5.9). Hold principal and maturity at their original values and consider a rate hike $R \in [R_0, R_0 + \bar{\Delta}]$, where $\bar{\Delta}$ is the

largest hike allowed by the deployability constraints of Section 4.6 (rate cuts are rarely optimal and are set aside here). Write $V(R) = NPV_{cond}(R) - C^{cap}(R)$ for the capital-adjusted value of the contract and $\bar{p}(R)$ for its prior-weighted acceptance probability, and define the two semi-elasticities

$$\varepsilon_V(R) \equiv \frac{V'(R)}{V(R)},$$
$$\varepsilon_p(R) \equiv -\frac{\bar{p}'(R)}{\bar{p}(R)} = \lambda\, PMT'(R) + \eta_i \sum_k \omega_k(R)\, m_k, \qquad \omega_k(R) = \frac{\pi_k\, p_k(R)}{\bar{p}(R)},$$

the percentage gain in contract value, and the percentage loss of acceptances, per percentage point of rate.

> **Proposition 1 (inverse semi-elasticity pricing rule).** On the region where $V(R) > 0$, the objective $W(R) = \bar{p}(R)\, V(R)$ satisfies $d \ln W / dR = \varepsilon_V(R) - \varepsilon_p(R)$. Hence: (i) a hike is profitable at the margin if and only if $\varepsilon_V(R_0) > \varepsilon_p(R_0) = \lambda\, PMT'(R_0) + \eta_i$; (ii) an interior optimum satisfies $\varepsilon_V(R^*) = \varepsilon_p(R^*)$; (iii) if $\varepsilon_V - \varepsilon_p$ crosses zero at most once from above on $[R_0, R_0 + \bar{\Delta}]$, the optimal rate is R_0 when $\varepsilon_V(R_0) \le \varepsilon_p(R_0)$, the cap $R_0 + \bar{\Delta}$ when $\varepsilon_V(R_0 + \bar{\Delta}) \ge \varepsilon_p(R_0 + \bar{\Delta})$, and the unique root otherwise; (iv) at the original contract,
>
> $$V'(R_0) = \Big[(1 - PD)\, PV'_{nd} + PD\, PV'_d\Big] - C^{cap\,\prime}(R_0) - (PV_{nd} - PV_d)\, PD\,(1 - PD) \left[\gamma_{emp} + \delta_{emp} \frac{PMT'(R_0)}{Y/12}\right],$$
>
> so, under the single-crossing condition of (iii), the set of borrowers whom the engine re-prices shrinks when γ_{emp}, δ_{emp} or η_i rises.

The proof is in Appendix D. Three implications carry into the results. First, part (i) is the rule a pricing desk would recognize, a Lerner condition written in semi-elasticities: raise the rate while one more point adds a larger share to the contract's value than it removes from its take-up. The Stiglitz–Weiss channel enters only through the bracketed term in (iv), which is why a larger γ_{emp} makes the engine *more* cautious. Second, because $\bar{p}$ is a mixture of exponentials, $\ln \bar{p}$ is convex: as the rate rises the price-sensitive type leaves first, ω_{high} falls and ε_p declines. Once a hike starts to pay, the incentive to continue therefore strengthens, and corner solutions at the cap are generic rather than exceptional. Third, for the illustrative median borrower of Figure 9 the rule can be evaluated directly: $\varepsilon_V(R_0) = 0.255$ against $\varepsilon_p(R_0) = 0.065$, so a hike pays; the unconstrained root lies at $\Delta R^* = +5.26$ points, beyond the five-point cap, where $\varepsilon_V = 0.068$ still exceeds $\varepsilon_p = 0.063$, so the constraint binds and the engine's Tier-1 offer of 18.1% is exactly the corner that part (iii) predicts.

4.9 Passive Regimes and the Three Profit Columns

Each passive PD model defines a regime that approves the *original* contract when its expected NPV is positive, $NPV_{cond} > 0$. Two stricter variants are reported for every model: **+NAM** approves only if $NPV_{cond} > CAC_i$, the same acquisition-margin logic the engine uses, and, for the Random Forest, **+NAM+K** additionally

charges the Basel capital cost, $NPV_{cond} - C^{cap} > CAC_i$. These variants matter for fairness: a comparison of the engine against a passive regime that is denied the NAM screen would attribute to *contract optimization* a gain that is really due to *acquisition discipline*. The best passive regime in every specification of this thesis is a +NAM variant.

Three profit columns are reported. *Expected* net profit uses model PDs (PD_{dyn} for engineered contracts) and is the headline measure, because it is the only column that reflects the default response to re-pricing. *Realized* net profit applies the observed default indicator to whichever contract is signed; for the engine this is an upper bound, because a borrower whose rate has been raised is assumed to default exactly as often as they did at the original rate, contradicting Table 1. *Pseudo-realized* net profit, new in this version, couples the observed outcome to the model's own re-pricing response: for a contract whose PD_{dyn} exceeds PD_{base}, an observed non-defaulter defaults with the incremental probability $(PD_{dyn} - PD_{base})/(1 - PD_{base})$, and symmetrically for contracts that lower PD. It is the honest backtest-like number for the engine and is reported alongside the headline throughout.

4.10 Vasicek Loss Simulation and Capital Definitions

For each of $N_{MC} = 10{,}000$ replications m a systematic factor $Z^{(m)} \sim N(0,1)$ is drawn, shared across all regimes so that their loss distributions are compared on common shocks. The 10,000-loan book is instantiated directly: the 2,000-borrower test book is replicated five-fold and each of the 10,000 slots draws its own type, tier and idiosyncratic default. Scaling a simulated 2,000-loan loss by five would be wrong, because it multiplies the idiosyncratic standard deviation by five, whereas in a genuine 10,000-loan book it grows only by $\sqrt{5}$; the direct simulation lets idiosyncratic risk diversify at the correct $\sqrt{N}$ rate, so that what survives into the 99.9% tail is predominantly systematic risk, the quantity Basel capital is meant to cover. Conditional on $Z^{(m)}$, each active loan i defaults with probability

$$PD_i^{cond}\big(Z^{(m)}\big) = \Phi\left(\frac{\Phi^{-1}(PD_i) - \sqrt{\rho(PD_i)}\, Z^{(m)}}{\sqrt{1-\rho(PD_i)}}\right),$$

where PD_i is the PD_{dyn} of the contract actually signed in that replication and $\rho(PD_i)$ is the Basel other-retail correlation. A Bernoulli default $D_i^{(m)} \sim \text{Bernoulli}\big(PD_i^{cond}(Z^{(m)})\big)$ is drawn and the portfolio loss is

$$\text{Loss}^{(m)} = \sum_{i \in \mathcal{A}^{(m)}} D_i^{(m)}\, EAD_i, \qquad EAD_i = LGD \cdot B_{\bar{\tau}-1}(c_i),$$

with $\mathcal{A}^{(m)}$ the set of active loans and c_i the signed contract. From the N_{MC} loss draws,

$$EL = \frac{1}{N_{MC}} \sum_{m} \text{Loss}^{(m)}, \qquad VaR_{99.9} = Q_{0.999}(\{\text{Loss}^{(m)}\}),$$
$$CVaR_{99.9} = \mathbb{E}[\text{Loss} \mid \text{Loss} \geq VaR_{99.9}], \qquad EC_{99.9} = VaR_{99.9} - EL.$$

The canonical definition of RAROC is (revenue − costs − EL)/EC. Because expected net profit already nets expected loss (the $PD \cdot PV_d$ term inside NPV_{cond}), subtracting EL again would count the credit loss twice, and the non-double-counting form is

$$RAROC = \frac{\text{expected corporate net profit}}{EC_{99.9}},$$

a *lifetime* measure over the 60-month horizon; the annualized equivalent is roughly one-fifth. The constant $\rho = 0.15$ of the proposal is reported as a stress case in Section 5.6.

4.11 Replication of the Frozen Prototype

The v13 code contains every feature as a switch. Setting all seven to their proposal values ($\bar{\tau} = 1$, $LGD = 1$, raw Random Forest PD, $\xi = 0$, no capital charge, $\rho = 0.15$, no constraints, one type, gate on the aggressive contract, Tier 2 may re-offer) reproduces the frozen `sidur_12.R` results exactly for the FICO and grade regimes (FICO: 5,550 approved, expected \$2,888,176, realized −\$1,926,891) and within the noise introduced by the sub-grade pooling for the others (sub-grade expected \$706,071 vs \$738,052; Random Forest \$2,639,571 vs \$2,554,867; engine \$5,736,050 with 5,015 approved vs \$5,534,248 with 4,943). The port is therefore validated, and every difference between the proposal's numbers and Section 5 is attributable to the seven features, whose individual contributions are decomposed in Section 5.12. All computations were run in R 4.5.1 (`sidur_13.R`); a Python reference implementation (`engine13/`) reproduces the same base-case ratio (1.440 before pooling in both languages) and was used for development and for the decomposition.

---

5. Empirical Results and Regulatory Risk Analysis

All figures are from the official run of `sidur_13.R` (R 4.5.1, seed 42, $N_{MC} = 10{,}000$), discounted at 5%, scaled to a 10,000-loan book and evaluated strictly out of sample. The base case is $\bar{\tau} = 12$, $LGD = 0.8$, Platt-calibrated Random Forest PD, all seven features on, and the Basel other-retail correlation in both the objective and the economic-capital simulation.

5.1 Master Summary Matrix

Table 2a. Out-of-sample master matrix, base case: volume and profit (10k-scaled, PV at 5%).

Regime	Approved	Rate	Capital ($)	Net **Exp** ($)	Net **Real** ($)	Net **Pseudo-real** ($)	Profit/$
FICO-only	8,615	86.2%	106,124,750	8,534,542	6,467,001	–	0.080
FICO-only +NAM	7,955	79.6%	97,756,250	**8,620,023**	6,389,942	–	0.088
Bank Grade	9,440	94.4%	115,784,500	7,495,017	6,148,100	–	0.065
Bank Grade +NAM	8,770	87.7%	105,484,875	7,539,657	6,557,554	–	0.071
Bank Sub-Grade	9,670	96.7%	118,759,125	7,375,943	6,212,643	–	0.062
Bank Sub-Grade +NAM	8,735	87.4%	105,703,125	7,472,670	6,380,922	–	0.071
Random Forest (Platt)	8,765	87.7%	108,743,875	8,399,550	6,588,538	–	0.077
Random Forest +NAM	8,045	80.5%	98,534,000	8,505,408	6,598,849	–	0.086
Random Forest +NAM+K	6,630	66.3%	80,188,625	8,182,840	5,456,918	–	0.102
Multi-Stage Engine	**8,632**	**86.3%**	**102,801,526**	**12,549,535**	14,634,889	**10,588,178**	**0.122**
Engine, unconstrained	8,875	88.8%	105,116,579	13,479,277	17,022,472	11,617,780	0.128
Engine, no capital charge	9,257	92.6%	111,514,367	12,539,275	15,263,837	10,850,283	0.112

Table 2b. Out-of-sample master matrix, base case: loss distribution and capital (10k-scaled, $N_{MC} = 10{,}000$).

Regime	EL ($)	$\text{VaR}_{99.9}$ ($)	$\text{CVaR}_{99.9}$ ($)	EC ($)	RAROC
FICO-only	11,983,874	23,920,963	25,130,626	11,937,089	71.5%
FICO-only +NAM	11,216,419	22,046,413	23,098,886	10,829,994	79.6%
Bank Grade	13,387,932	26,320,136	27,002,392	12,932,203	58.0%
Bank Grade +NAM	12,729,393	24,590,428	25,844,588	11,861,035	63.6%
Bank Sub-Grade	13,550,882	26,695,097	27,609,282	13,144,216	56.1%
Bank Sub-Grade +NAM	12,899,748	24,728,727	25,891,423	11,828,979	63.2%
Random Forest (Platt)	12,115,374	23,989,188	25,053,675	11,873,814	70.7%
Random Forest +NAM	11,206,047	22,285,876	23,003,268	11,079,830	76.8%
Random Forest +NAM+K	9,195,222	18,046,631	18,591,776	8,851,409	92.5%
Multi-Stage Engine	14,427,366	26,994,722	27,945,536	12,567,355	**99.9%**
Engine, unconstrained	15,581,528	28,775,260	29,657,211	13,193,732	102.2%
Engine, no capital charge	15,717,302	29,210,286	30,322,085	13,492,984	92.9%

Monte-Carlo 95% interval for the engine's expected net profit: [$12,508,451, $12,590,548]. The engine's realized column is an upper bound (Section 4.9); the pseudo-realized column is the comparable backtest.

Three results carry the thesis. First, the engine's expected net profit of **$12.55M** is **1.456×** the best passive regime's $8.62M, and its pseudo-realized profit of $10.59M is still 1.228× that figure, so the advantage does not depend on trusting the model's own PD. Second, the advantage is not bought with volume or with risk. The engine approves 86.3% of applicants, between the FICO regime (86.2%) and its +NAM variant (79.6%); deploys $102.8M of capital, less than every full-volume bank scorecard; and earns **0.122 per dollar deployed** against 0.088 for the best passive. Its economic capital ($12.57M) is within $0.7M of the FICO regime's, so its lifetime **RAROC of 99.9%** exceeds the best passive's 79.6% by twenty points and the Random Forest's

70.7% by twenty-nine (Figure 1, Figure 8). Third, the two features that constrain the engine cost it money in exactly the way they should: removing the deployability constraints adds $0.93M (7.4%) of expected profit by permitting hikes above five points and payment burdens above 15%, and removing the capital charge leaves expected profit essentially unchanged (−$10k) while adding 625 approvals, $8.7M of capital and $0.9M of economic capital, cutting RAROC from 99.9% to 92.9%. The capital charge, in other words, is not a cost to profit but a filter on capital consumption.

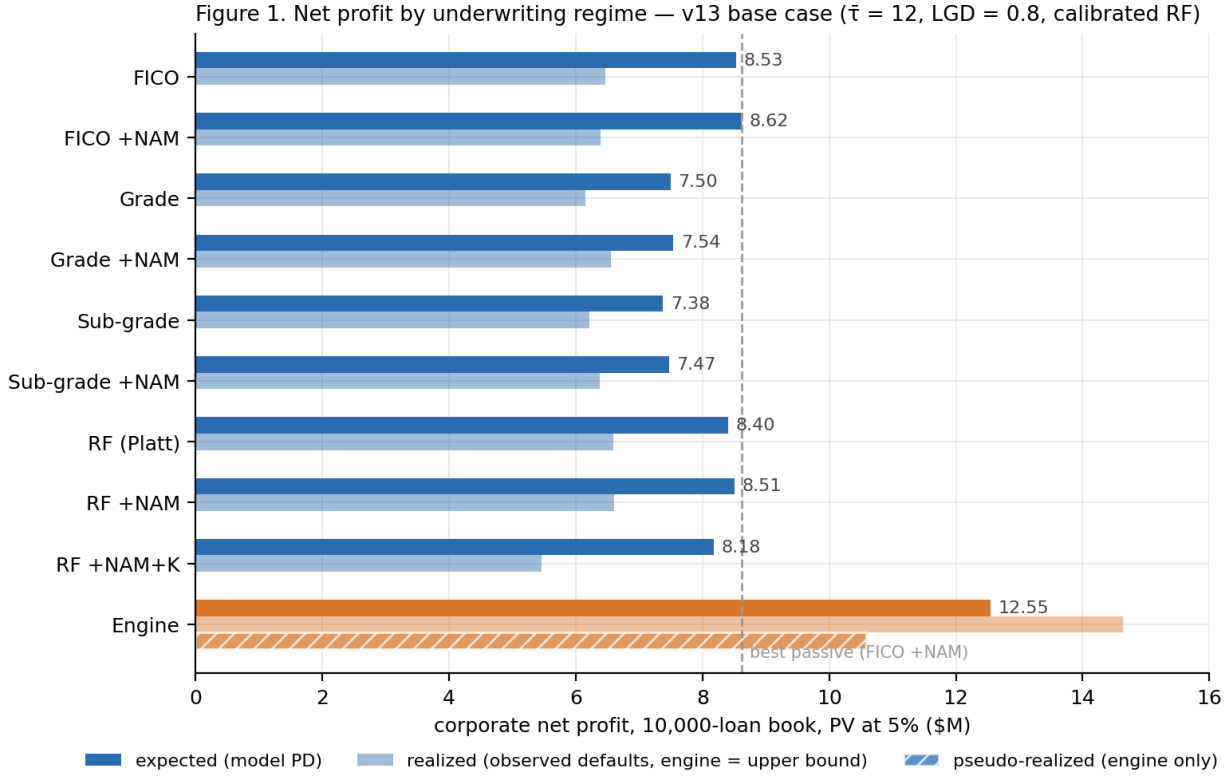


Figure 1. Net profit by underwriting regime, base case: expected (model PD), realized (observed defaults; an upper bound for the engine) and pseudo-realized (engine only).

Among the passive regimes two regularities matter for what follows. Adding the acquisition-margin screen (+NAM) raises every regime's profit and RAROC, so the fair comparator for the engine is always a +NAM regime. And the two bank scorecards that approve almost everyone (grade 94.4%, sub-grade 96.7%) earn the *least*: at this loss profile nearly every original contract has positive expected NPV, so a passive regime's only lever, the reject decision, has little to act on.

5.2 Statistical Significance of the Engine's Advantage

The comparison in Table 2a is a difference between two sums over the same 2,000 test borrowers, so its sampling uncertainty can be measured directly. For each borrower we record the engine's expected net contribution and that of a passive regime (zero if the regime rejects), and resample borrowers with replacement 5,000 times,

holding every estimated model and every contract fixed (Efron & Tibshirani, 1993). The design is paired: each resample draws the same borrowers for both regimes, so the uncertainty that the regimes share cancels.

Table 3. Paired bootstrap of the engine's advantage over the four strongest passive regimes (10k-scaled; $B = 5{,}000$ resamples of the test borrowers).

Comparator	Difference ($)	95% interval ($)	Ratio	95% interval	Engine higher
FICO-only +NAM (best passive)	3,929,512	3,578,309 – 4,270,746	1.456	1.403 – 1.513	68.5%
Random Forest +NAM	4,044,127	3,885,971 – 4,203,409	1.475	1.449 – 1.505	86.2%
Random Forest +NAM+K	4,366,695	4,192,603 – 4,541,017	1.534	1.500 – 1.571	86.2%
Bank Sub-Grade +NAM	5,076,865	4,775,309 – 5,370,853	1.679	1.630 – 1.730	76.4%

In none of the 5,000 resamples was the engine's advantage zero or negative against any comparator, so the one-sided bootstrap p-value is below $1/5{,}000$ in every row; paired t-statistics on the borrower-level differences are 22.19, 49.44, 48.12 and 32.82 respectively. The last column counts borrowers whose expected contribution under the engine strictly exceeds that under the comparator; ties (borrowers rejected by both) are counted against the engine.

The advantage over the best passive regime is \$3.93M with a 95% interval of \$3.58M to \$4.27M, and the ratio's interval, 1.40 to 1.51, lies far from one. The intervals are narrowest against the Random Forest regimes, which share the engine's PD model and therefore its sampling noise, and widest against the FICO regime, which does not. The bootstrap measures uncertainty over which borrowers happen to be in the test set, taking the models as given; uncertainty over the models themselves is the subject of the re-estimation exercise in Section 5.11, where the engine wins all twenty independent partitions, an outcome whose probability under the null that engine and best passive regime are equally likely to win any split is $0.5^{20} \approx 9.5 \times 10^{-7}$. On both notions of sampling variation, H1 is supported.

5.3 Discrimination versus Value

Table 4. Out-of-sample AUC and value by regime.

Score used to rank borrowers	OOS AUC	Regime	Net expected ($)	RAROC
Integrated logistic rating (ceiling, not a regime)	**0.6691**	–	–	–
Sub-grade PD	0.6597	Sub-grade +NAM	7,472,670	63.2%
Random Forest PD (Platt)	0.6497	RF +NAM	8,505,408	76.8%
Grade PD	0.6419	Grade +NAM	7,539,657	63.6%
FICO PD	0.6285	FICO +NAM	**8,620,023**	79.6%

Score used to rank borrowers	OOS AUC	Regime	Net expected ($)	RAROC
Engine profit score $\bar{p}\,(NPV - C^{cap})$	**0.5607**	**Engine**	**12,549,535**	**99.9%**

Within the passive regimes the relation between discrimination and value is flat to negative: the sub-grade model has the highest AUC of the four and the lowest profit, the FICO model the lowest AUC and the highest profit (Figure 2). The reason is that at a realistic loss profile all four PD models approve most of the book, and their expected profit is then dominated by which *marginal* borrowers they reject, a property of the PD *level* near the decision boundary, not of the ranking. The engine's own profit score has, by construction, the lowest default-AUC in the study (0.56): it ranks borrowers by acceptance-weighted, capital-adjusted value, in which a risky borrower who accepts a well-priced contract outranks a safe borrower who would walk. That is Hand's (2009) point in an operating system: the quantity that matters to a lender is not the one that AUC measures.

Are the AUC differences among the passive models themselves meaningful? Table 4b applies DeLong, DeLong and Clarke-Pearson's (1988) test for correlated ROC curves to every pair.

Table 4b. DeLong tests of paired AUC differences (test set, $n = 2{,}000$**).**

Model A	Model B	AUC A	AUC B	z	p
Sub-grade	FICO	0.6597	0.6285	2.10	0.035
Grade	FICO	0.6419	0.6285	0.91	0.365
Random Forest (Platt)	FICO	0.6497	0.6285	1.16	0.246
Sub-grade	Grade	0.6597	0.6419	2.70	0.007
Random Forest (Platt)	Sub-grade	0.6497	0.6597	–0.69	0.491
Integrated logit	Sub-grade	0.6691	0.6597	1.08	0.282
Integrated logit	Random Forest (Platt)	0.6691	0.6497	1.50	0.135

Most of the differences are within sampling error: the Random Forest does not significantly out-discriminate even the FICO-only score ($p = 0.25$), and the integrated logistic rating, the best discriminator in the study, is not significantly better than the sub-grade or the Random Forest. The one clearly significant ordering is the sharpest illustration of H2. The sub-grade model discriminates significantly better than FICO ($z = 2.10$, $p = 0.035$), yet its best regime earns $7.47M against FICO's $8.62M, $1.15M less. A statistically established advantage in ranking coexists with a lower profit, because profit at the decision margin depends on the *level* of the predicted PD relative to each contract's breakeven PD, where the sub-grade logit's coarse, step-shaped predictions approve more loans that are only marginally profitable. We therefore state the paradox in its defensible form. Ranking quality is neither necessary nor sufficient for value; a classifier used as a filter leaves the pricing, structuring and capital levers unused, and in this data those levers are worth about 45% of the best passive regime's profit.

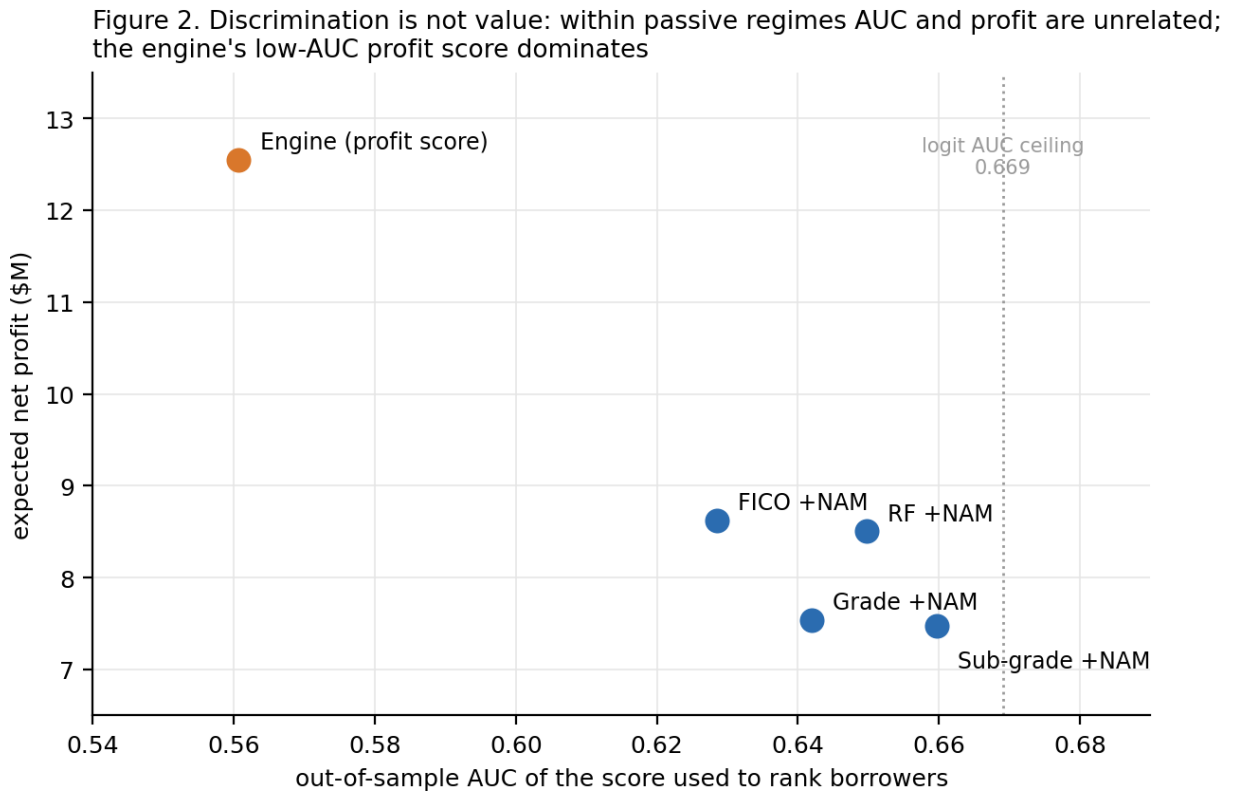


Figure 2. Discrimination versus value: out-of-sample AUC of the score used to rank borrowers against expected net profit.

Likelihood: a test of levels rather than ranks. The AUC is blind to exactly the property that the formal argument of Section 1.2 identifies as decisive, the level of the predicted PD. The statistic that does measure levels is the likelihood. For a binary outcome the out-of-sample log-likelihood $\ell(p) = \sum_i [D_i \ln p_i + (1 - D_i) \ln(1 - p_i)]$ is maximized in expectation only by the true default probabilities, so it rewards correct levels as well as correct ordering. Table 4c evaluates every PD model on the 2,000 test outcomes by this criterion and sets beside it the profit that each PD earns.

Table 4c. Likelihood-based comparison of the PD models (test set, $n = 2{,}000$) and the profit each PD earns (10k-scaled).

PD model	OOS log-likelihood	Log-loss	OOS pseudo-R^2	AUC	Passive +NAM net ($)	Engine net with this PD ($)
Integrated logistic rating	**−858.1**	**0.4291**	**0.054**	**0.6691**	7,919,108	12,342,416
Sub-grade	−863.7	0.4318	0.048	0.6597	7,472,670	–
Random Forest, Platt (used by the engine)	−870.6	0.4353	0.041	0.6497	8,505,408	**12,549,535**
Grade	−871.2	0.4356	0.040	0.6419	7,539,657	–
Random Forest, raw	−876.4	0.4382	0.034	0.6497	6,607,158	10,379,497

PD model	OOS log-likelihood	Log-loss	OOS pseudo-R^2	AUC	Passive +NAM net ($)	Engine net with this PD ($)
FICO-only	−879.7	0.4398	0.031	0.6285	**8,620,023**	–

Probabilities are clipped to $[10^{-4}, 1-10^{-4}]$*, as inside the engine. Log-loss is* $-\ell/n$*; the pseudo-*R^2 *is* $1-\ell(p)/\ell(\bar{D})$*, measured against a constant equal to the training default rate. The integrated logistic rating is not one of the thesis's lending regimes; its passive column applies the +NAM rule to its PD for comparison. On the training sample AIC ranks the parametric scorecards integrated logit (6,825), grade (6,861), sub-grade (6,870), FICO (6,940), while BIC, with its heavier penalty on the 56-parameter logit, reverses the top: grade (6,903), FICO (6,954), sub-grade (7,072), integrated logit (7,182). The Random Forest has no likelihood-based parameter count.*

Table 4d. Vuong tests of paired out-of-sample log-likelihood differences (test set, $n = 2{,}000$**).**

Model A	Model B	ℓ A	ℓ B	z	p
Random Forest (Platt)	FICO	−870.6	−879.7	1.37	0.169
Random Forest (Platt)	Sub-grade	−870.6	−863.7	−1.11	0.266
Random Forest (Platt)	Integrated logit	−870.6	−858.1	−2.00	0.046
Random Forest (Platt)	Random Forest (raw)	−870.6	−876.4	0.94	0.349
Sub-grade	FICO	−863.7	−879.7	2.80	0.005
Integrated logit	Sub-grade	−858.1	−863.7	1.27	0.204
Integrated logit	FICO	−858.1	−879.7	3.06	0.002

$z = \sqrt{n}\,\bar{d}/s_d$*, where* d_i *is the difference between the two models' log-likelihood contributions for test borrower* i *(Vuong, 1989). Both models are estimated on the training set and evaluated on held-out data, so no penalty for the number of parameters is applied.*

Three results follow. First, likelihood does no better than AUC at ordering passive profit. The FICO-only model has the lowest likelihood of the six and earns the highest passive profit; the integrated logistic rating has the highest likelihood and, used as a passive screen, would earn $0.70M less than FICO; across the six PD models the rank correlation between likelihood and passive profit is −0.14. The reason is the one given by the formal argument: likelihood, like AUC, is an average over the whole book, whereas passive profit is decided by the borrowers whose PD lies near their breakeven, so a model can fit the book well on average and still misplace those borrowers. Second, most likelihood differences are within sampling error. The calibrated Random Forest is not distinguishable from the FICO score ($p = 0.17$), from the sub-grade model ($p = 0.27$) or even from its own raw version ($p = 0.35$), and it is only marginally below the integrated logit ($p = 0.046$); the clear differences are that the sub-grade model and the integrated logit fit better than FICO ($p = 0.005$ and 0.002). Even for the engine, the PD with the best likelihood is not the one that earns most ($12.34M with the integrated logit against $12.55M with the calibrated forest), although the two are close. Third, and most telling,

Platt scaling, which changes neither the ranking nor, significantly, the likelihood, changes profit by \$1.90M for the passive regime and by \$2.17M for the engine: what matters is the level of PD at the decision margin, which neither summary statistic isolates. Neither ranking quality nor global fit is a sufficient statistic for value. Both describe the input, and the choice among PD models must be made, as in this thesis, in the currency of the decision. This extends H2 from discrimination to fit, and it is why the evaluation in Section 5 is conducted in profit and capital rather than in AUC or likelihood.

5.4 What the Engine Does, and to Whom

Table 5. Engine internals, base case (test book, $n = 2{,}000$).

Quantity	Value
Borrowers passing the NAM gate	1,732 (86.6%)
Group 1 (positive original NPV) sourced	1,710
Group 2 sourced / expected rescued	22 / 16.4
Mean Tier-1 rate shift, Group 1	+3.27 pts
Share of Tier-1 offers with a hike > 0.25 pts	85.4%
Share of Tier-1 offers at 36 months	0.0%
Mean Tier-1 principal fraction	0.974
Mean Tier-1 acceptance probability	0.797
Route probabilities $P_1/P_2/P_3$ (Group 1 mean)	0.662 / 0.238 / 0.099
Gross expected profit / CAC paid (10k-scaled)	\$14,793,905 / \$2,244,370
Expected mechanism value net of capital and CAC	\$8,716,916

The engine is, in practice, a *rate-setting* mechanism. It almost never cuts principal (mean fraction 0.974) and never shortens maturity, because both raise the payment-to-income ratio and, through $\delta_{emp} = 6.575$, the default probability by more than they save. It raises the rate on 85% of Group-1 borrowers by 3.27 points on average, and the distribution of hikes is bimodal (Figure 10): a third of the offers sit exactly at the five-point cap, another 14% at zero, and the rest spread between. Two-thirds of Group-1 borrowers accept the Tier-1 offer, a quarter accept the counter-offer, and one in ten retain the original contract. The illustrative median borrower of Figure 9 (\$10,000 at 13.1%, PD 0.16) receives a Tier-1 offer at 18.1%, which raises the modelled PD to 0.236 and is accepted with probability 0.725; a rejection shifts the lender's belief toward the price-sensitive type and produces a counter-offer at 17.6%.

Two features of this behavior deserve comment. The concentration at the cap says that the constraint binds for a third of the book: a lender willing to hike further would earn more (Section 5.9 shows the unconstrained gain is modest, 7%, because the demand and default responses flatten the objective). The absence of term compression is a genuine finding about the affordability channel, not an artifact of the grid: the 36-month option is available to every borrower and is never chosen once the payment-to-income response is priced.

Who bears the re-pricing? Table 6 breaks the engine's treatment down by the platform's credit grade and by income quartile, and compares the engine's expected net contribution with that of the best passive regime for

the same borrowers.

Table 6. Heterogeneity of the engine's treatment, base case (test book; net profit 10k-scaled).

Group	n	Mean PD	Sourced	Mean Tier-1 hike (pts)	At the +5 cap	Tier-1 acceptance	Engine net ($)	FICO +NAM net ($)
Grade A	364	0.105	49.7%	+3.65	0.0%	0.721	695,523	39,537
Grade B	786	0.148	90.8%	+4.80	76.2%	0.711	4,612,877	1,944,192
Grade C	490	0.178	97.8%	+3.03	8.1%	0.820	3,623,278	2,839,333
Grade D	297	0.212	99.7%	+0.46	0.0%	0.969	2,755,816	2,877,223
Grade E	45	0.216	97.8%	0.00	0.0%	1.000	668,170	653,764
Grades F–G	18	0.263	100.0%	–0.11	0.0%	1.000	193,872	265,975
Income Q1 (lowest)	500	0.194	82.4%	+3.06	30.3%	0.819	1,706,036	1,630,157
Income Q2	561	0.160	84.3%	+3.32	38.1%	0.803	3,230,569	2,146,124
Income Q3	443	0.147	88.7%	+3.38	35.8%	0.788	3,265,422	2,012,671
Income Q4 (highest)	496	0.136	91.5%	+3.33	31.9%	0.779	4,347,507	2,831,071

Hikes, cap shares and acceptances are means over Group-1 borrowers who pass the NAM gate. Grade A hikes are bounded by the rule that the new rate may not exceed 1.5 times the original.

The pattern is the one Proposition 1 predicts, and it is not the one a critic of risk-based re-pricing would fear. The engine re-prices the middle of the credit distribution and leaves the riskiest borrowers alone: grade-B borrowers receive the largest hikes (+4.80 points, three-quarters of them at the cap), grade-D borrowers almost none (+0.46), and grades E–G none at all. For high-PD borrowers the Stiglitz–Weiss term $PD(1-PD)(PV_{nd}-PV_d)\gamma_{emp}$ in part (iv) of the proposition is large and the capital charge is steep, so a further point of rate destroys more value than it adds; for the safest borrowers the hike aversion η_i is highest and the 1.5× rate rule binds. The value of the mechanism is correspondingly concentrated: grades A–C account for $4.11M of the engine's $3.93M advantage over the best passive regime, while in grades D–G the engine earns $0.18M *less* than the FICO +NAM regime, because it declines to source a few marginal borrowers that the passive rule accepts. Across income quartiles the treatment is flat, with mean hikes between +3.06 and +3.38 points; the lowest-income quartile receives the smallest hike and gains least from the mechanism, and the engine's advantage rises with income. Section 6.2 returns to what this implies for fair lending.

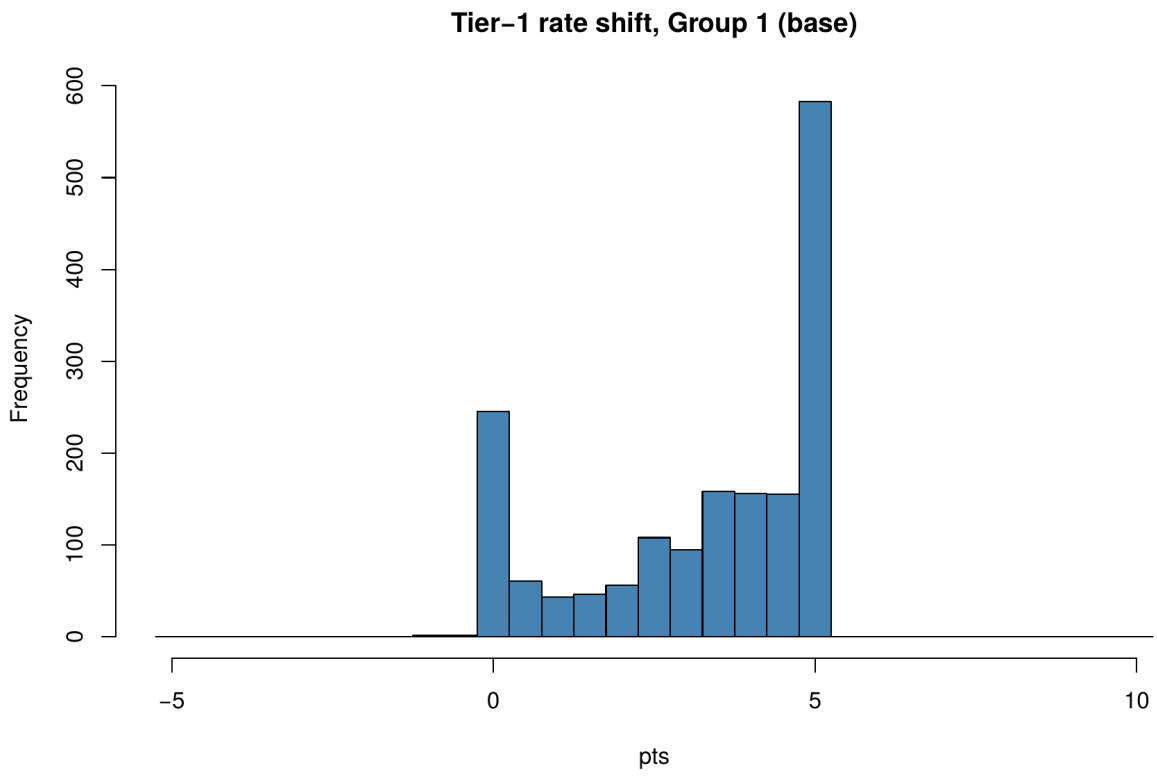


Figure 10. Distribution of Tier-1 rate shifts for Group-1 borrowers (base case; from `sidur_13_plots.pdf`). The mass at +5 points is the deployability cap.

5.5 Which Contract Dimensions Create the Value

The engine searches over three axes, but Section 5.4 suggests that it uses essentially one. Table 7 measures the value of each axis directly, by re-running the engine with the grid restricted to subsets of the three dimensions while every other feature, including the NAM gate and the capital charge, is left on.

Table 7. Axis decomposition: engine expected net profit with the contract grid restricted (10k-scaled).

Axes the engine may move	Engine expected ($)	Pseudo-realized ($)	Ratio to best passive	Approved	Mean Tier-1 hike	Mean principal fraction	36-month share
None (original contract only)	8,182,840	5,456,918	0.949	6,630	0.00	1.000	0%
Principal and term	8,233,695	5,616,559	0.955	6,815	0.00	0.990	0%
Rate only	12,301,606	10,244,505	1.427	8,342	+3.14	1.000	0%
Rate and term	12,301,606	10,244,505	1.427	8,342	+3.14	1.000	0%
Rate and principal	12,549,535	10,588,178	1.456	8,632	+3.27	0.974	0%
All three (base case)	12,549,535	10,588,178	1.456	8,632	+3.27	0.974	0%

Three results follow. First, with no contract axis at all the engine reduces exactly to a passive regime: its expected profit of \$8,182,840 is, to the dollar, that of the Random Forest +NAM+K regime, because the mechanism value of an unchanged contract is its capital-adjusted NPV and the gate then applies the same rule. This is an internal consistency check on the implementation, and it shows that the selection machinery alone (gate, capital charge, calibrated PD) does not beat the best passive regime; the value comes from changing contracts. Second, the rate axis alone delivers \$4.12M of the \$4.37M that the full engine adds over this gate-only benchmark, 94.3%; allowing principal cuts adds the remaining \$0.25M (5.7%); and the term axis adds nothing, with or without the rate. Third, principal and term together, without the rate, are worth \$0.05M. H3 is supported in a sharp form: in this testbed the engine's advantage is a pricing advantage. That is a result about the estimated affordability response ($\delta_{emp} = 6.575$), which makes any payment-raising change expensive, and about a demand side that rewards lower payments only weakly; it is not a statement that principal and maturity are irrelevant levers in general, and a lender whose borrowers value longer maturities, as Attanasio et al. (2008) find, or whose grid allowed maturities above 60 months, could find value on axes that are inactive here.

Figure 11. The engine's advantage is a pricing advantage, concentrated in the middle of the credit distribution

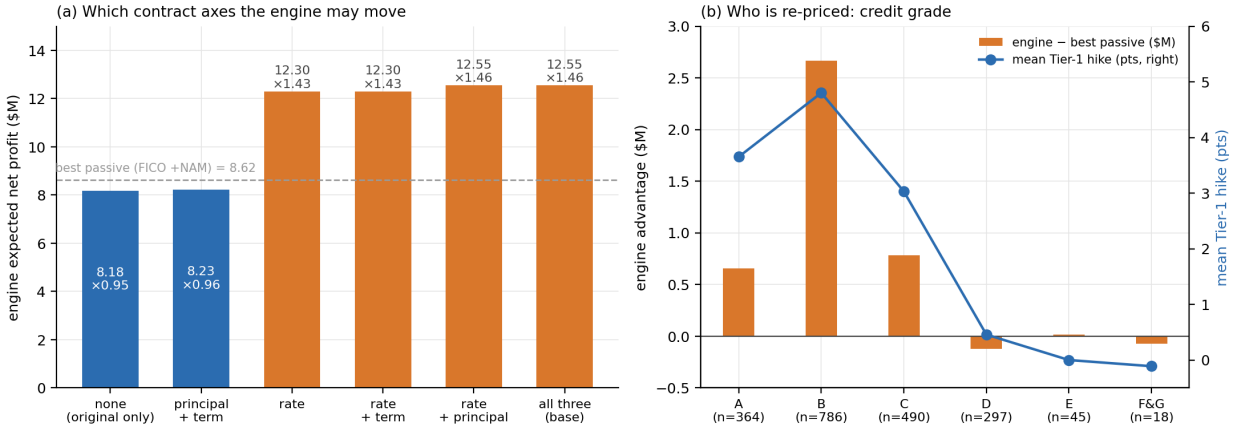


Figure 11. (a) Axis decomposition: engine expected net profit when the grid is restricted to subsets of the three contract dimensions. (b) Heterogeneity by credit grade: mean Tier-1 hike and the engine's advantage over the best passive regime.

Is the grid fine enough? The engine chooses among 1,178 contracts per borrower, with the principal in steps of 5% and the rate in steps of half a point. Its choice is optimal on that grid by construction; the question is how much a finer search would add. Table 7b re-solves the full engine on grids two and five times finer in each of these two dimensions, up to 27,482 candidate contracts per borrower. The coarser grids are nested in the finer ones, so every borrower's best Tier-1 value can only improve.

Table 7b. Optimality gap of the contract grid: the engine re-solved on finer grids (10k-scaled).

Grid (principal step, rate step)	Contracts per borrower	Engine expected ($)	Gain over the base grid	Ratio to best passive	Mean Tier-1 value gain, Group 1	Group-1 borrowers gaining > 1%	Mean Tier-1 hike
5%, 0.5 pt (base)	1,178	12,549,535	–	1.456	–	–	+3.27
2.5%, 0.25 pt	4,514	12,572,552	+0.18%	1.459	0.59%	11.0%	+3.30
1%, 0.1 pt	27,482	12,582,491	+0.26%	1.460	0.84%	11.3%	+3.31

The term axis is kept at 36 and 60 months, the maturities offered on the platform. The Tier-1 value gain is the relative improvement of the borrower's best Tier-1 objective $\bar{p}\,(NPV - C^{cap})$ *over the base grid, averaged over Group-1 borrowers who pass the NAM gate.*

A 23-fold refinement of the grid raises the engine's expected profit by $32,956, or 0.26%, and moves its ratio to the best passive regime from 1.456 to 1.460. The gains shrink as the grid is refined (0.18% for the first step, a further 0.08% for the second), and they are concentrated: for the median Group-1 borrower the best Tier-1 value improves by less than 0.01%, for one borrower in nine by more than 1%, and at the 95th percentile by 6.8%. The 1,178-point grid is therefore within about a quarter of a percent of the best contracts the model admits at much finer resolution, and its coarseness is neither a source of the engine's advantage nor a material cost to it. This is the precise and defensible sense in which the engine's choice is the best available option: to within 0.3%, it is the optimum of the stated objective over the feasible contract space, given the estimated default and demand responses. It is not a claim that no other policy could do better under a different model of the borrower; that question is the subject of Sections 5.8 to 5.10.

5.6 Basel III/IV Risk and Capital Efficiency

The engine's headline regulatory result is a lifetime RAROC of **99.9%**, against 79.6% for the best passive regime and 92.5% for the Random Forest with both the NAM and capital screens (Table 2b). The comparison with the latter is instructive: the RF +NAM+K regime achieves its RAROC by shrinking, approving 66% of the book and deploying $80M, while the engine achieves a higher RAROC at 86% approval and $103M deployed. Capital-aware *pricing* dominates capital-aware *rejection* because it keeps the borrower and charges for the capital instead of forgoing both.

Table 8. Economic capital under the Basel other-retail correlation (base) and under a constant $\rho = 0.15$ (stress), 10k-scaled.

Regime	EC, $\rho(PD)$ ($)	RAROC	EC, $\rho = 0.15$ ($)	RAROC
FICO-only +NAM	10,829,994	79.6%	29,521,389	29.2%
Bank Sub-Grade +NAM	11,828,979	63.2%	31,862,958	23.5%
Random Forest +NAM	11,079,830	76.8%	29,186,841	29.1%
Random Forest +NAM+K	8,851,409	92.5%	23,600,755	34.7%
Multi-Stage Engine	**12,567,355**	**99.9%**	**32,309,604**	**38.8%**

Regime	EC, $\rho(PD)$ ($)	RAROC	EC, $\rho = 0.15$ ($)	RAROC

The mortgage-level correlation triples every regime's economic capital and compresses every RAROC, but preserves the ordering: the engine's 38.8% still exceeds the best passive's 29.2% and the RF +NAM+K regime's 34.7%. The engine's economic capital is the highest in absolute terms ($12.6M vs $10.8M for FICO +NAM) because its book is both larger and, by design, riskier on the intensive margin: re-priced borrowers carry higher PD_{dyn}. What the capital charge inside the objective guarantees is that this additional risk is paid for, which is why the ratio, not the level, is the relevant comparison.

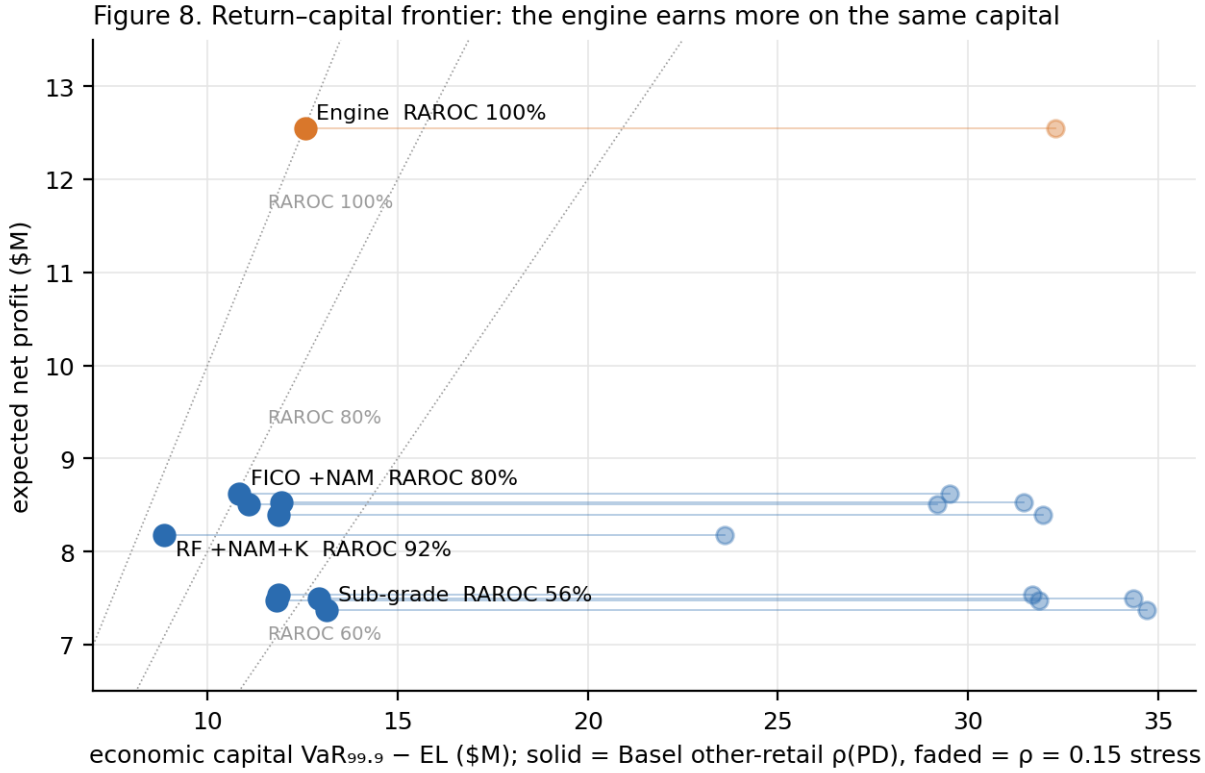


Figure 8. Return–capital frontier: expected net profit against economic capital under the Basel other-retail correlation (solid) and under ρ = 0.15 (faded). Dotted rays are constant-RAROC lines.

5.7 Loss-Profile Scenarios and the Conservative Bound

The two most influential assumptions in the model are the time to default $\bar{\tau}$ and the loss given default. Table 9 varies them jointly from the proposal's immediate-total-loss specification to a lenient profile.

Table 9. Engine versus best passive regime across loss profiles (expected net, 10k-scaled).

$\bar{\tau}$ (months)	LGD	Engine expected ($)	Engine pseudo-real ($)	Engine approval	Best passive ($)	Ratio
1	1.0 (proposal spec)	4,150,391	1,812,221	41.3%	3,075,125 (RF +NAM)	1.350
6	1.0	5,901,282	3,190,662	53.0%	4,298,802 (FICO +NAM)	1.373
12	1.0	8,650,662	6,060,455	68.2%	6,100,480 (FICO +NAM)	1.418
12	**0.8 (base)**	**12,549,535**	**10,588,178**	**86.3%**	**8,620,023 (FICO +NAM)**	**1.456**
18	0.7	17,521,011	16,146,084	94.6%	11,797,798 (FICO +NAM)	1.485
24	0.6	22,194,201	21,167,310	98.3%	14,742,409 (FICO +NAM)	1.505

Levels move by a factor of five across the grid; the ratio moves between 1.35 and 1.51 and the engine wins in every cell (Figure 5). The proposal's specification is the harshest: with every default an immediate total loss, only 41% of the book is worth writing under any regime, the passive scorecards' *realized* profit is negative (FICO −$1.93M, grade −$2.60M, sub-grade −$0.81M; only RF +NAM+K realizes a gain, +$0.52M), and the engine's realized profit of $2.86M is the only substantially positive one. That is the result the proposal reported, and it is retained here as a **conservative bound**: an engine that dominates when every default costs the whole principal on day one dominates a fortiori under any realistic profile. But it should not be read as the base case. Under the base profile all passive regimes are profitable, and the earlier claims that passive scoring "destroys value" and that the engine's realized profit exceeding its expected profit demonstrates a "conservative" risk model are both artifacts of $\bar{\tau} = 1$, $LGD = 1$; the second is additionally an artifact of the realized column's failure to re-price default (Section 4.9), which is why this thesis reports the pseudo-realized column instead.

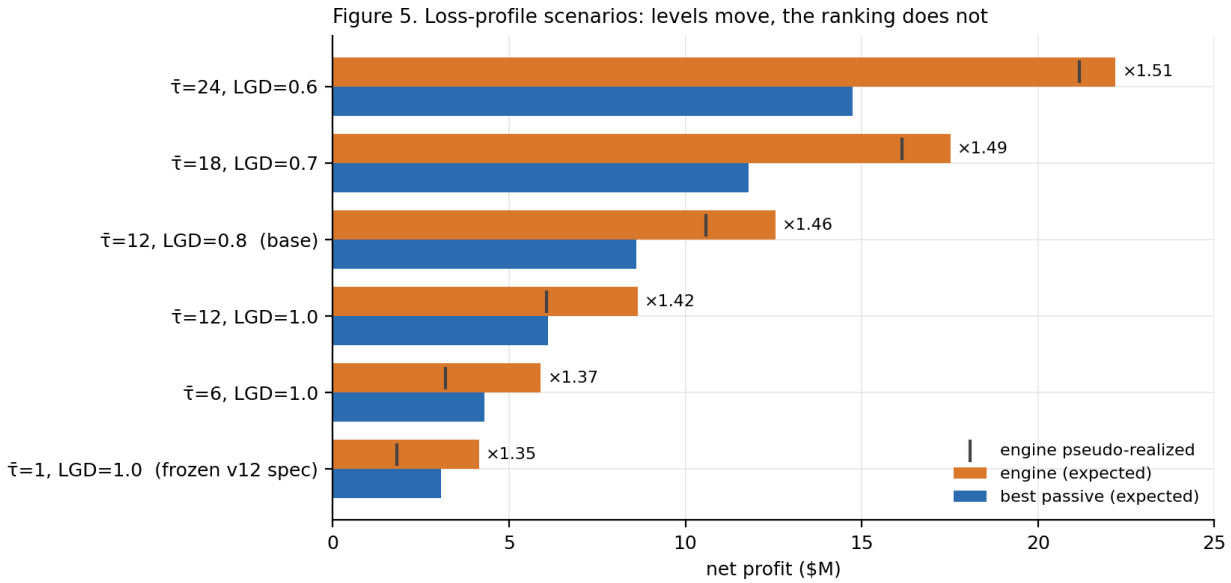


Figure 5. Loss-profile scenarios: engine and best passive expected net profit, with the engine's pseudo-realized profit marked.

5.8 Tornado Sensitivity

Table 10. Sensitivity of engine expected net profit to ±25% parameter shocks (base $12,549,535).

Parameter	Net at −25% ($)	Net at +25% ($)	Swing ($)
LGD	16,631,617	8,650,662	7,980,956
r_{disc} (cost of funds)	16,184,742	9,101,787	7,082,956
γ_{emp} (rate → PD)	14,290,710	10,989,553	3,301,156
$\bar{\tau}$ (months to default)	10,988,070	14,077,425	3,089,355
κ (CAC base rate)	13,165,531	11,871,825	1,293,705
η_0 (hike aversion)	12,847,976	12,261,044	586,933
h (capital hurdle)	12,575,028	12,372,701	202,327
λ (payment aversion)	12,641,374	12,453,942	187,432
δ_{emp} (PTI → PD)	12,616,229	12,480,889	135,340
ξ (elasticity heterogeneity)	12,568,250	12,513,449	54,801
ϕ (CAC size elasticity)	12,574,152	12,520,408	53,744
α (principal demand elasticity)	12,573,558	12,522,864	50,694
γ_{cac} (CAC rate aversion)	12,544,947	12,547,479	2,531
η_{cut} (cut attraction)	12,549,662	12,549,010	652

The tornado (Figure 3) separates the fourteen parameters into three groups. The *loss profile and cost of funds* (LGD, r_{disc}, $\bar{\tau}$) dominate, with swings of $3–8M, but these are book-level assumptions that move every regime together and leave the ranking intact (Table 9). The *empirical Stiglitz-Weiss elasticity* γ_{emp} is next ($3.3M): a stronger default response to price makes hikes costlier, so the identification caveat of Section 3.2, which suggests $\hat{\gamma}_{emp}$ is an upper bound, is conservative for the engine. The *behavioral and cost parameters* (κ, η_0, h,

λ, δ_{emp}, ξ, ϕ, α, γ_{cac}, η_{cut}) together move profit by less than \$1.3M each, and the last six by under \$0.2M. In the proposal κ was the dominant sensitivity; once the loss profile is modelled explicitly it is fifth, and the parameters that matter are the ones that a lender can observe (its cost of funds and its recovery experience) rather than the ones that are calibrated.

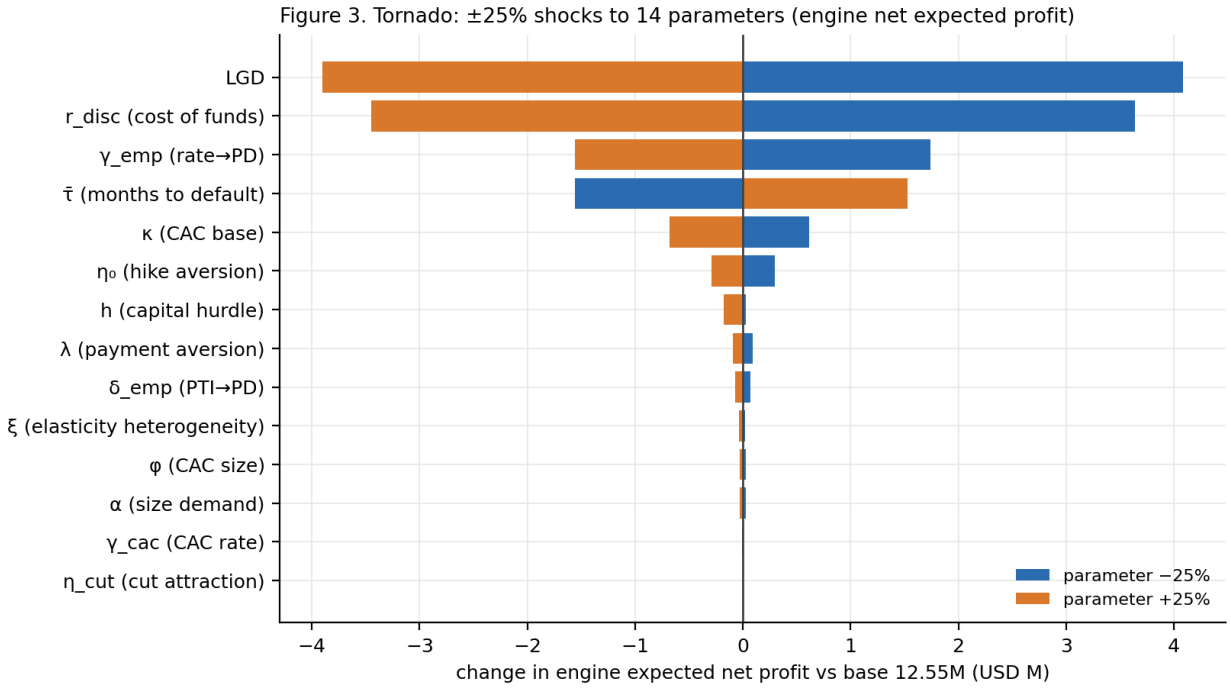


Figure 3. Tornado: change in engine expected net profit under ±25% shocks to fourteen parameters.

5.9 Breakeven Elasticity and Mechanism Sensitivities

The demand side is calibrated, so the natural question is how wrong the calibration would have to be for the conclusion to change. Table 11 raises the hike-aversion parameter from a quarter of its base value to ten times it.

Table 11. Engine versus best passive as a function of hike aversion η_0.

η_0	Price elasticity at the mean rate	Engine expected (\$)	Ratio to best passive	Mean Tier-1 hike (pts)
0.0125	−0.17	13,390,850	1.553	+4.30
0.025	−0.33	13,131,788	1.523	+4.00
0.05 (base)	**−0.66**	**12,549,535**	**1.456**	**+3.27**
0.1	−1.32	11,427,597	1.326	+2.00
0.2	−2.64	10,262,528	1.191	+0.83
0.3	−3.96	9,659,361	1.121	+0.38
0.5	−6.60	9,063,563	1.051	+0.07

The engine's edge declines smoothly with demand elasticity and never closes within the range examined (Figure 4). At $\eta_0 = 0.5$, a price elasticity of −6.6 at the mean rate, roughly ten times the elasticities reported for consumer credit, the engine has stopped hiking rates almost entirely (mean +0.07 points) and still earns 5% more than the best passive regime, entirely from the NAM gate, the capital-aware selection and the occasional principal or rate adjustment. The edge is therefore a rate-setting edge at plausible elasticities and a selection edge at implausible ones.

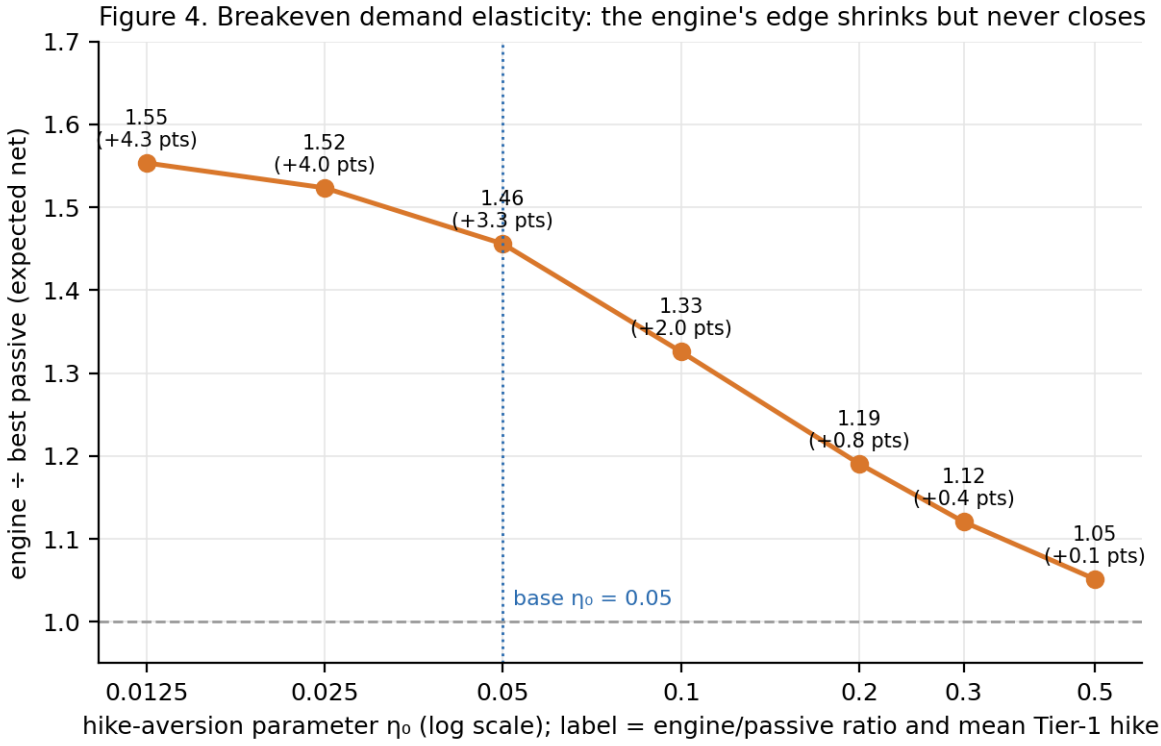


Figure 4. Breakeven demand elasticity: engine ÷ best passive as a function of the hike-aversion parameter η_0 (labels: ratio and mean Tier-1 hike).

Table 12. Mechanism and deployability sensitivities (ratio of engine to best passive expected net).

Parameter	Values	Ratio
Attrition after a rejected Tier-1 offer ω	0 / 0.10 / 0.25 / 0.50	1.456 / 1.427 / 1.385 / 1.315
Elasticity heterogeneity ξ	0 / 0.25 / 0.50 / 1.0	1.466 / 1.461 / 1.456 / 1.436
Capital hurdle h	0 / 0.05 / 0.10 / 0.15 / 0.20	1.455 / 1.460 / 1.456 / 1.413 / 1.345
Maximum rate hike (pts)	2.5 / 5 / 7.5 / 10	1.290 / 1.456 / 1.467 / 1.467

Attrition is the one mechanism parameter with material leverage: if half the borrowers who decline the aggressive offer leave rather than hear the counter-offer, the ratio falls from 1.46 to 1.31, because the Tier-2 route carries a quarter of Group-1 profit. Raising the hurdle to 20% lowers the ratio to 1.35 by making capital

expensive for everyone; tightening the hike cap to 2.5 points lowers it to 1.29, while loosening it beyond 7.5 points adds nothing, because the demand and default responses make hikes above that level unattractive on their own (Figure 7).

Figure 7. Mechanism and deployability sensitivities (ratio of expected net profit)

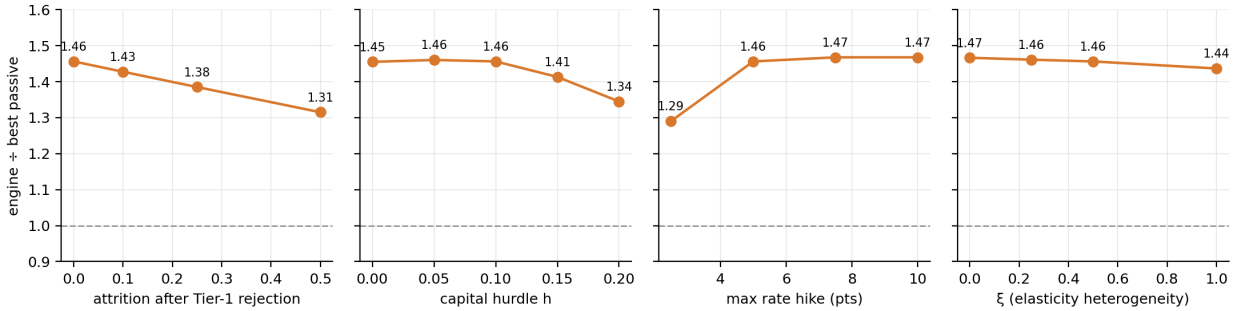


Figure 7. Mechanism and deployability sensitivities: engine ÷ best passive under attrition, capital hurdle, maximum hike and elasticity heterogeneity.

5.10 Acquisition Cost, the PD Model, and Identification of the Rate Channel

Three remaining assumptions could, in principle, manufacture the engine's advantage: the acquisition-cost model, the choice of PD model, and the identification of the rate elasticity of default. Table 13 relaxes each in turn.

Table 13. Robustness of the engine's advantage (expected net profit, 10k-scaled).

Specification	Engine ($)	Best passive ($)	Ratio	Engine approval
Base case	12,549,535	8,620,023 (FICO +NAM)	1.456	86.3%
(a) Acquisition cost				
Zero acquisition cost for all regimes ($\kappa = 0$)	15,072,627	10,759,508 (FICO)	1.401	92.0%
(b) PD model feeding the engine and the Random Forest regimes				
Random Forest, raw (uncalibrated)	10,379,497	8,620,023 (FICO +NAM)	1.204	73.5%
Integrated logistic rating	12,342,416	8,620,023 (FICO +NAM)	1.432	93.8%
(c) Rate elasticity of default				
Pooled estimate, $\gamma_{emp} = 0.0898$ (s.e. 0.0104) (base)	12,549,535	8,620,023	1.456	86.3%
No rate channel, $\gamma_{emp} = 0$	19,399,305	8,620,023	2.250	–
Sub-grade fixed effects, $\gamma_{emp} = -0.1349$ (s.e. 0.1038)	26,450,146	8,620,023	3.068	–

In (a) the NAM screen is inactive for every regime, since $CAC = 0$. In (b) the passive benchmark is unchanged

because the best passive regime is FICO-based. In (c) the Stiglitz–Weiss regression of Table 1 is re-estimated with sub-grade fixed effects on the training set; the affordability coefficient is stable ($\delta_{emp} = 6.247$, s.e. 1.102, against 6.575).

(a) Acquisition cost. The proposal's largest sensitivity was the marketing base rate. Removing acquisition cost entirely raises every regime's profit and switches off the NAM screen, yet the engine still earns 1.40 times the best passive regime, and its absolute advantage *grows*, from \$3.93M to \$4.31M. The advantage is not an artifact of charging passive regimes for marketing they cannot avoid.

(b) The PD model. Feeding the engine the integrated logistic rating instead of the calibrated Random Forest leaves the ratio at 1.43, so the result does not depend on the machine-learning model; the engine is a decision layer on top of whatever PD model a lender uses. Feeding it the *uncalibrated* Random Forest lowers the engine's profit by \$2.17M and the ratio to 1.20, because the raw forest over-predicts default (mean PD 0.186 against an observed 0.169, Table 1b) and the optimizer duly under-prices and under-sources. This is the operational meaning of Section 2.5's claim that calibration, not discrimination, is what a contract optimizer needs: calibration alone is worth more than 15% of the engine's profit.

(c) Identification of the rate channel. Section 3.2 argued that the pooled estimate $\hat{\gamma}_{emp} = 0.0898$ confounds a behavioral response with risk-based pricing. Adding sub-grade fixed effects removes the between-grade component: the sub-grade explains 99.5% of the variance of the contractual rate in the training data, and the rate varies by only 0.34 points on average within a sub-grade, against 3.94 points overall. On that thin within-grade variation the rate coefficient is not distinguishable from zero (-0.135, 95% interval $[-0.338,\ 0.069]$), while the affordability coefficient is essentially unchanged. The interval excludes the pooled estimate, which supports reading $\hat{\gamma}_{emp}$ as an upper bound on the behavioral channel. The fixed-effects point estimate is not credible as a causal effect either, since the residual within-grade variation reflects the timing of the platform's pricing changes, and we do not use it; but the bound matters for the conclusion. If the true rate response lies anywhere below the pooled estimate, the engine's advantage is larger than reported: with no rate channel at all it would be 2.25 times the best passive regime. The base case is therefore conservative with respect to the one parameter that most directly disciplines rate-setting, and H4 is supported.

Likelihood-ratio evidence on the two channels. The same conclusion follows from the likelihood of the default model itself. Table 13b tests the Stiglitz–Weiss terms with likelihood-ratio tests of the regression of Table 1 against restricted versions without them, in the pooled sample and with sub-grade fixed effects, and reports the out-of-sample gain in log-likelihood of the unrestricted model on the test set.

Table 13b. Likelihood-ratio tests of the Stiglitz–Weiss channels in the default logit (training set; out-of-sample gain on the test set).

Specification	Terms dropped	LR	df	p	ΔAIC	ΔBIC	OOS log-likelihood gain	Vuong z
Pooled	rate and PTI	133.39	2	1.1×10^{-29}	−129.4	−115.4	+20.32	3.50

Specification	Terms dropped	LR	df	p	ΔAIC	ΔBIC	OOS log-likelihood gain	Vuong z
Pooled	rate	72.63	1	1.6×10^{-17}	−70.6	−63.6	+10.40	2.45
Pooled	PTI	36.09	1	1.9×10^{-9}	−34.1	−27.1	+5.82	1.88
Sub-grade fixed effects	rate and PTI	34.03	2	4.1×10^{-8}	−30.0	−16.1	+5.02	1.67
Sub-grade fixed effects	rate	1.67	1	0.196	+0.3	+7.3	−0.43	−0.67
Sub-grade fixed effects	PTI	32.02	1	1.5×10^{-8}	−30.0	−23.0	+5.49	1.86

ΔAIC and ΔBIC are the unrestricted minus the restricted model, so negative values favour keeping the terms. The out-of-sample gain is the test-set log-likelihood of the unrestricted model minus that of the restricted model, both estimated on the training set; Vuong's z is computed from the per-borrower differences.

In the pooled data both channels are overwhelmingly significant and both improve out-of-sample fit. Within sub-grades the picture splits. The payment-burden channel survives intact: dropping it costs a likelihood ratio of 32.0 ($p = 1.5 \times 10^{-8}$), both information criteria retain it, and it improves out-of-sample fit by 5.5 log-points. The rate channel does not: its likelihood ratio falls from 72.6 to 1.67 ($p = 0.20$), both AIC and BIC prefer to drop it, and out-of-sample fit is marginally better without it. The likelihood therefore confirms, with a test designed for nested models, what the confidence interval of Table 13 showed: the affordability response is identified from variation within credit grades, while the rate response in the pooled regression largely reflects the platform's pricing schedule. Since a smaller rate response only enlarges the engine's advantage, the base case remains conservative.

5.11 Split Stability

A single 80/20 partition can flatter a model that was designed while looking at it. The entire pipeline, including the Random Forest, its out-of-fold calibration, the four scorecards, the Stiglitz-Weiss regression and the engine, was therefore re-estimated on twenty alternative random partitions (seeds 0–19), analytically without the Monte-Carlo step. Table 14 summarizes; Appendix A lists every split.

Table 14. Twenty random re-partitions: expected net profit (10k-scaled) and the engine's ratio to the best passive regime.

Regime	Mean ($)	Std. dev. ($)	2.5% ($)	97.5% ($)
FICO-only +NAM	8,141,128	245,149	7,716,568	8,499,586
Random Forest +NAM	8,076,722	234,419	7,689,975	8,453,496
Random Forest +NAM+K	7,755,836	248,239	7,352,809	8,154,610
Bank Grade +NAM	7,248,530	200,860	7,003,550	7,610,768
Bank Sub-Grade +NAM	7,092,925	209,346	6,738,450	7,407,254

Regime	Mean ($)	Std. dev. ($)	2.5% ($)	97.5% ($)
Multi-Stage Engine	**11,978,093**	**398,506**	**11,338,770**	**12,679,179**
Engine, pseudo-realized	10,820,293	–	–	–
Engine ÷ best passive	**1.466**	–	min **1.375**	max **1.549**

The engine wins every one of the twenty splits, with a ratio between 1.375 and 1.549 (Figure 6). The canonical seed-42 result ($12.55M, ratio 1.456) lies inside the split distribution and slightly above its mean, so the headline is not a favorable draw. The identity of the best passive regime, by contrast, is unstable: FICO +NAM in fourteen splits and Random Forest +NAM in six, with the two never more than $0.5M apart. The passive regimes are statistically indistinguishable from one another; the engine is distinguishable from all of them at every split. This is also the appropriate answer to the concern that the seven design choices were tuned to the canonical partition: every choice is either estimated on the training fold only or held fixed across all twenty partitions, and the conclusion survives all of them.

Figure 6. Split stability: the engine wins every one of 20 random partitions

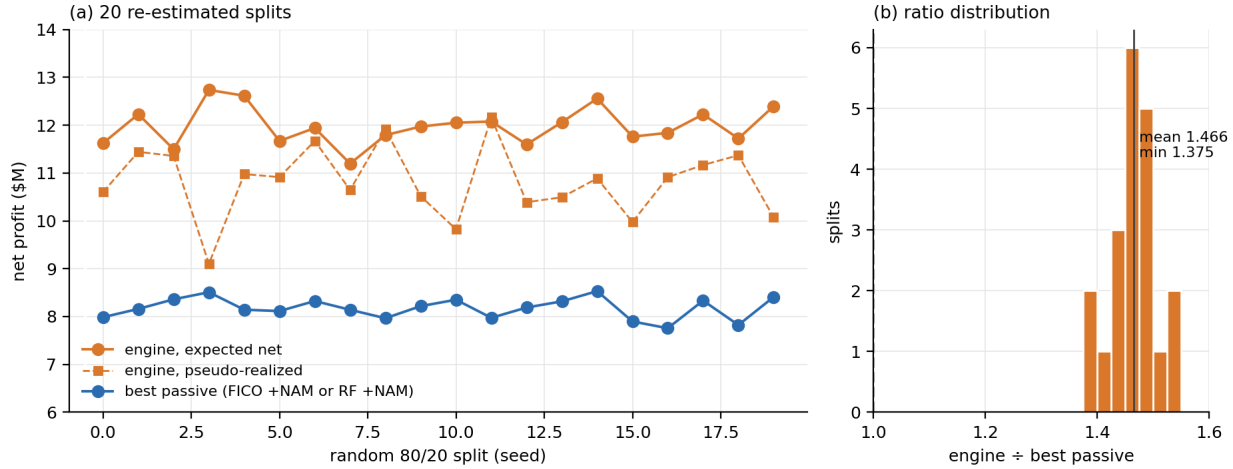


Figure 6. Split stability: engine and best passive expected net profit across twenty random partitions, and the distribution of the ratio.

5.12 Decomposition: What Each Feature Contributes

The seven features were introduced cumulatively in the Python reference implementation, in the order listed in Table 15, on the same seed-42 partition (before the sub-grade pooling; the Python and R base-case ratios agree to three decimals). The table reports the engine's ratio to the best passive regime after each step and is the decomposition of the gap between the proposal's headline (2.08×) and this thesis's (1.44× before pooling, 1.456× after).

Table 15. Fix ladder (Python reference implementation, seed 42, $\rho_{EC} = 0.15$ held fixed for comparability).

Step	Engine expected ($)	Engine approval	Best passive ($)	Ratio	Mean hike (pts)
S0 Proposal specification (`sidur_12`)	6,215,950	54.9%	2,990,410	2.079	+3.51
+F1 Loss profile ($\bar{\tau} = 12$, LGD 0.8)	13,001,968	85.5%	7,952,630	1.635	+3.89
+F4 Gate on mechanism value	13,006,899	86.5%	7,952,630	1.636	+3.86
+F2 Calibrated PD (Platt)	12,827,866	91.5%	8,000,834	1.603	+3.48
+F3 Heterogeneous elasticity ($\xi = 0.5$)	12,135,546	91.3%	8,000,834	1.517	+3.11
+F5 Capital charge in objective	12,231,934	84.9%	8,000,834	1.529	+3.69
+F7 Deployability constraints	11,521,702	83.1%	8,000,834	1.440	+2.91
+F6 Two types, Bayesian Tier 2	11,518,092	83.0%	8,000,834	1.440	+2.96

The loss profile accounts for most of the change: it raises every regime's profit and, because passive regimes gain proportionally more from being allowed to keep a defaulted borrower's early payments, it narrows the ratio from 2.08 to 1.64. Calibration and heterogeneous elasticity remove a further 0.12, the deployability constraints 0.09; the capital charge slightly *raises* the ratio by removing capital-inefficient approvals; the Bayesian Tier 2 is value-neutral at zero attrition (its role is to make the mechanism coherent, and it matters once attrition is positive). The engine's advantage in the proposal was therefore overstated by roughly 40%, and the corrected advantage of about 45% is what remains after every feature that a skeptical reviewer would ask for has been added.

---

6. Managerial Implications and Policy Discussion

6.1 Implementation for Banks and Fintech Lenders

The framework unifies three functions that are typically siloed, risk, pricing and marketing, and adds a fourth, capital, that is usually applied after the fact. The NAM gate integrates the acquisition budget into the credit decision so that media spend is committed only against a mechanism whose expected value covers it; the bargaining tree replaces a single offer with a governed menu whose second step learns from the first; the capital charge makes every offer pay for the capital it consumes; and the deployability constraints keep the resulting offers inside the range a consumer-protection review would accept. The starkest managerial finding is not that passive scoring loses money, which under a realistic loss profile it does not, but that a lender who uses a machine-learning score only to accept and reject is forgoing roughly a third of the value that the same score, used to price and structure, would create. Because every quantity is estimated on held-out data, re-estimated on twenty partitions and stress-tested parameter by parameter, the pipeline supports the model-risk governance that supervisors expect.

Carrying the framework from the testbed to a lender's own book requires five inputs, all of which a regulated lender either already holds or can collect. (1) A *calibrated* PD for each applicant from whatever model the lender uses; the engine consumes the level of PD, so calibration on the lender's own outcomes matters more

than the choice of algorithm. (2) The lender's own *loss profile*: recovery rates and the timing of default by product, which replace the assumed $\bar{\tau}$ and LGD, the two most influential parameters in the model (Section 5.8). (3) The *default response* to contract terms, γ_{emp} and δ_{emp}, ideally identified from past pricing changes or randomized price tests rather than from cross-sectional variation (Section 3.2). (4) *Offer-and-response data* to estimate the acceptance function, which turns the breakeven analysis of Section 5.9 into a point estimate; a lender can generate such data at low cost with small randomized price variations, as Karlan and Zinman (2009) did in the field. (5) Its *cost of funds, capital hurdle and acquisition costs*, which enter as parameters. With these inputs the same code runs unchanged; without (3) and (4), the conservative defaults of this thesis and the breakeven analysis indicate how much of the estimated advantage survives.

6.2 Fair Lending and Disparate Impact

Active re-pricing raises consumer-protection concerns under the Equal Credit Opportunity Act and analogous regimes, because observationally similar borrowers may receive different offers, and Fuster et al. (2022) show that better statistical technology in credit markets can widen outcome gaps across groups even when it raises average efficiency. Four design features bound the risk. The Tier-3 fallback guarantees that no borrower is made worse off than under their original contract; the endogenous PD uses repayment-relevant covariates only; the deployability constraints cap every hike at five points and 1.5 times the original rate and prevent any offer from raising a borrower's payment burden above 15% of income; and the heterogeneous-elasticity specification makes the mechanism hike less where borrowers have more alternatives.

The heterogeneity analysis of Section 5.4 adds evidence that the proposal lacked, and it points in a reassuring direction. The engine does not concentrate its hikes on the riskiest or the poorest borrowers. Grades D–G, the highest-PD fifth of the book, receive essentially no hike, because the Stiglitz–Weiss response and the capital charge make re-pricing them value-destroying (Proposition 1). The lowest-income quartile receives the smallest mean hike of the four. The re-pricing falls on mid-grade borrowers with moderate risk and moderate price sensitivity. This is an outcome of the model's economics, not a constraint imposed on it, and it depends on the estimated responses; a lender with a weaker default response to price would see hikes move down the credit distribution. Grade and income are also not protected characteristics, and the data contain none. A formal disparate-impact audit on protected classes and their proxies, of the kind Fuster et al. (2022) perform, therefore remains a prerequisite for any deployment.

6.3 Aligning Marketing and Capital with Underwriting

Two operational insights follow from the sensitivities. The marketing base rate κ was the proposal's dominant sensitivity and is now fifth; what dominates is the lender's cost of funds and recovery experience. This reframes the priority for a lender adopting the framework: measuring LGD and time-to-default on its own book is worth more than refining the marketing model. And the comparison between capital-aware pricing (the engine, RAROC 99.9% at 86% approval) and capital-aware rejection (RF +NAM+K, 92.5% at 66% approval) shows that the capital constraint is better handled by charging for capital in the price than by refusing the borrower: the

former keeps the customer and the return; the latter forgoes both. Finally, the PD-model robustness of Section 5.10 gives a lender a clear priority when upgrading its risk models: calibrating the probability it already has is worth more than switching algorithms. Replacing the calibrated Random Forest with a logistic rating costs 1.6% of the engine's profit, while dropping calibration costs more than 17%.

---

7. Conclusion, Limitations, and Future Research

7.1 Summary of Findings

This thesis asked how a lender should act on a risk signal when the borrower's default and acceptance both respond to the terms offered. The answer is an active mechanism that prices, structures and gates every contract jointly under an empirically estimated Stiglitz–Weiss default response, an explicit demand response, a realistic loss profile and a Basel capital charge. Its optimum obeys a simple inverse semi-elasticity rule, raising the rate while one more point adds more to the contract's value than it removes from its take-up, and the rule explains both when the engine re-prices and why its hikes cluster at the cap. Each of the four hypotheses is supported on the testbed.

H1 (value). The engine earns \$12.55M against \$8.62M for the best of nine passive regimes, an advantage of \$3.93M with a paired-bootstrap interval of \$3.58M to \$4.27M. Its lifetime RAROC is 99.9% against 79.6%, at a comparable approval rate and on comparable capital, and it wins all twenty re-estimated partitions ($p \approx 10^{-6}$). Re-solving the engine on a grid 23 times finer adds 0.26%, so its choice is, to within that margin, the optimum of its objective over the contract space.

H2 (discrimination is not value). Among passive regimes AUC does not order profit. The one statistically significant AUC advantage, sub-grade over FICO, comes with \$1.15M *less* profit, and the engine's own ranking has the lowest AUC in the study. Global fit tells the same story: the PD model with the lowest likelihood (FICO) earns the most passive profit, and Platt scaling, which leaves the ranking unchanged, moves the engine's profit by \$2.17M.

H3 (mechanism). The advantage is a pricing advantage: the rate axis delivers 94% of it, principal cuts 6% and maturity nothing. It shrinks smoothly with demand elasticity but does not close even at ten times the base value. It is concentrated on mid-grade borrowers, as Proposition 1 predicts.

H4 (robustness). The ranking survives six loss profiles, a mortgage-level asset correlation, zero acquisition cost, an alternative PD model and fourteen parameter shocks. It is conservative with respect to the rate elasticity of default, which is not identified within credit grades. Likelihood-ratio tests confirm that the affordability channel is identified within credit grades and the rate channel is not.

The Machine-Learning Underwriting Paradox survives in its defensible form. Better ranking is neither necessary nor sufficient for better lending: a lender that uses a calibrated score only as a filter leaves roughly a third of the attainable value unexploited, and the lever that recovers it is price.

7.2 Limitations

Several choices bound the interpretation. The *identification* of the rate elasticity $\widehat{\gamma}_{emp}$ is not clean: rates are risk-based-priced, the coefficient blends a behavioral response with a pricing correlation, and within credit sub-grades there is too little rate variation to identify it (Section 5.10); it is best read as an upper bound, which is conservative for the engine. The *loss profile* is assumed, not estimated, because the data contain neither default timing nor recoveries; $\bar{\tau}$ and LGD are the two most influential parameters in the model, and their effect is reported as a grid rather than a point. The *demand side* is calibrated rather than estimated, which is why the breakeven analysis, not the base case, is the appropriate statement of the demand-side result. The *Platt calibration* over-corrects on the test set (slope 1.26): a limitation of the five-fold out-of-fold procedure on 8,000 observations that compresses the PD differences the optimizer uses and therefore biases the engine's expected profit downward. The *pooling* of five rare sub-grades into one level, required for the repeated-split analysis, moves the base case by about one percent. The realized backtest cannot observe how borrowers would have defaulted at the re-priced terms, so the pseudo-realized column is a model-coupled construction, not an observation. The *single-factor* ASRF model with the Basel correlation captures one systematic factor; a multi-factor or stochastic-correlation extension would thicken the tail. Prepayment and refinancing, which would remove value from the top of the balance solid, are not modelled. The data contain no origination dates, so *out-of-time* validation, the test that credit models most often fail, is impossible here; the twenty-split exercise establishes stability across borrowers, not across time. The evidence comes from one testbed population of unsecured, fixed-rate marketplace loans; applying the framework to other products or lenders requires re-estimating its inputs on their data (Section 6.1), and the magnitude of the advantage, though not the mechanism, should be expected to differ. Finally, the analysis is *partial equilibrium*: the borrower's outside option is held fixed, whereas if every lender adopted capital-aware re-pricing, competition would compress the margins the engine exploits, and the estimated advantage should be read as the value to a single lender of adopting the mechanism first, not as a market-wide gain.

7.3 Future Research

Natural extensions include an out-of-time validation on dated vintages; estimation of the loss profile (default timing and recovery) from payment histories, which would replace the two most influential assumptions with data; instrumental-variable or discontinuity-based identification of the adverse-selection elasticity; estimation of the acceptance function from offer-and-response funnel data, which would turn the breakeven analysis into a point estimate; a formal solution of the optimal screening menu with a continuum of types, benchmarked against the two-type Bayesian tree; a prepayment model; a disparate-impact audit; and a multi-factor macro model with recession scenarios, in which the geometric picture of Section 4.3 would show the guillotine moving with the cycle.

References

Adams, W., Einav, L., & Levin, J. (2009). Liquidity constraints and imperfect information in subprime lending. *American Economic Review, 99*(1), 49–84. https://doi.org/10.1257/aer.99.1.49

Attanasio, O. P., Goldberg, P. K., & Kyriazidou, E. (2008). Credit constraints in the market for consumer durables: Evidence from micro data on car loans. *International Economic Review, 49*(2), 401–436. https://doi.org/10.1111/j.1468-2354.2008.00485.x

Basel Committee on Banking Supervision. (2005). *An explanatory note on the Basel II IRB risk weight functions*. Bank for International Settlements.

Basel Committee on Banking Supervision. (2019). *The Basel Framework, CRE31: IRB approach — risk weight functions* (version published 15 December 2019). Bank for International Settlements.

Berg, T., Burg, V., Gombović, A., & Puri, M. (2020). On the rise of FinTechs: Credit scoring using digital footprints. *The Review of Financial Studies, 33*(7), 2845–2897. https://doi.org/10.1093/rfs/hhz099

Breiman, L. (2001). Random forests. *Machine Learning, 45*(1), 5–32.

Chen, P.-F., Lo, S., & Tang, H.-Y. (2022). What if borrowers stop paying their loans? Investors' rates of return on a peer-to-peer lending platform. *International Review of Economics & Finance, 77*, 359–377. https://doi.org/10.1016/j.iref.2021.10.011

Croux, C., Jagtiani, J., Korivi, T., & Vulanovic, M. (2020). Important factors determining Fintech loan default: Evidence from a LendingClub consumer platform. *Journal of Economic Behavior & Organization, 173*, 270–296. https://doi.org/10.1016/j.jebo.2020.03.016

DeLong, E. R., DeLong, D. M., & Clarke-Pearson, D. L. (1988). Comparing the areas under two or more correlated receiver operating characteristic curves: A nonparametric approach. *Biometrics, 44*(3), 837–845. https://doi.org/10.2307/2531595

Dobbie, W., & Skiba, P. M. (2013). Information asymmetries in consumer credit markets: Evidence from payday lending. *American Economic Journal: Applied Economics, 5*(4), 256–282. https://doi.org/10.1257/app.5.4.256

Efron, B., & Tibshirani, R. J. (1993). *An introduction to the bootstrap*. Chapman & Hall. https://doi.org/10.1201/9780429246593

Einav, L., Jenkins, M., & Levin, J. (2012). Contract pricing in consumer credit markets. *Econometrica, 80*(4), 1387–1432. https://doi.org/10.3982/ECTA7677

Fuster, A., Goldsmith-Pinkham, P., Ramadorai, T., & Walther, A. (2022). Predictably unequal? The effects of machine learning on credit markets. *The Journal of Finance, 77*(1), 5–47. https://doi.org/10.1111/jofi.13090

Gordy, M. B. (2003). A risk-factor model foundation for ratings-based bank capital rules. *Journal of Financial Intermediation, 12*(3), 199–232. https://doi.org/10.1016/S1042-9573(03)00040-8

Hand, D. J. (2009). Measuring classifier performance: A coherent alternative to the area under the ROC curve. *Machine Learning, 77*(1), 103–123. https://doi.org/10.1007/s10994-009-5119-5

Jagtiani, J., & Lemieux, C. (2019). The roles of alternative data and machine learning in fintech lending: Evidence from the LendingClub consumer platform. *Financial Management, 48*(4), 1009–1029. https://doi.org/10.1111/fima.12295

Karlan, D., & Zinman, J. (2009). Observing unobservables: Identifying information asymmetries with a consumer credit field experiment. *Econometrica, 77*(6), 1993–2008. https://doi.org/10.3982/ECTA5781

Khandani, A. E., Kim, A. J., & Lo, A. W. (2010). Consumer credit-risk models via machine-learning algorithms. *Journal of Banking & Finance, 34*(11), 2767–2787.

LendingClub Corporation. (2017). *Annual report on Form 10-K for the fiscal year ended December 31, 2016*. U.S. Securities and Exchange Commission.

Lessmann, S., Baesens, B., Seow, H.-V., & Thomas, L. C. (2015). Benchmarking state-of-the-art classification algorithms for credit scoring: An update of research. *European Journal of Operational Research, 247*(1), 124–136. https://doi.org/10.1016/j.ejor.2015.05.030

Merton, R. C. (1974). On the pricing of corporate debt: The risk structure of interest rates. *The Journal of Finance, 29*(2), 449–470.

Mussa, M., & Rosen, S. (1978). Monopoly and product quality. *Journal of Economic Theory, 18*(2), 301–317.

Niculescu-Mizil, A., & Caruana, R. (2005). Predicting good probabilities with supervised learning. In *Proceedings of the 22nd International Conference on Machine Learning* (pp. 625–632). Association for Computing Machinery. https://doi.org/10.1145/1102351.1102430

Phillips, R. L. (2005). *Pricing and revenue optimization*. Stanford University Press.

Platt, J. C. (1999). Probabilistic outputs for support vector machines and comparisons to regularized likelihood methods. In A. J. Smola, P. Bartlett, B. Schölkopf, & D. Schuurmans (Eds.), *Advances in large margin classifiers* (pp. 61–74). MIT Press.

Rochet, J.-C., & Choné, P. (1998). Ironing, sweeping, and multidimensional screening. *Econometrica, 66*(4), 783–826.

Serrano-Cinca, C., & Gutiérrez-Nieto, B. (2016). The use of profit scoring as an alternative to credit scoring systems in peer-to-peer (P2P) lending. *Decision Support Systems, 89*, 113–122. https://doi.org/10.1016/j.dss.2016.06.014

Stiglitz, J. E., & Weiss, A. (1981). Credit rationing in markets with imperfect information. *The American Economic Review, 71*(3), 393–410.

Vasicek, O. A. (2002). Loan portfolio value. *Risk, 15*(12), 160–162.

Verbraken, T., Bravo, C., Weber, R., & Baesens, B. (2014). Development and application of consumer credit

scoring models using profit-based classification measures. *European Journal of Operational Research, 238*(2), 505–513. https://doi.org/10.1016/j.ejor.2014.04.001

Vuong, Q. H. (1989). Likelihood ratio tests for model selection and non-nested hypotheses. *Econometrica, 57*(2), 307–333. https://doi.org/10.2307/1912557

Zhang, J., & Thomas, L. C. (2012). Comparisons of linear regression and survival analysis using single and mixture distributions approaches in modelling LGD. *International Journal of Forecasting, 28*(1), 204–215. https://doi.org/10.1016/j.ijforecast.2010.06.002

Appendix A. Repeated Random Splits (seeds 0–19)

Seed	Engine expected ($)	Engine pseudo-real ($)	Best passive regime	Best passive ($)	Ratio
0	11,623,680	10,601,693	Random Forest +NAM	7,986,792	1.455
1	12,229,717	11,442,935	FICO-only +NAM	8,159,852	1.499
2	11,496,612	11,357,426	FICO-only +NAM	8,360,155	1.375
3	12,736,532	9,101,167	FICO-only +NAM	8,508,790	1.497
4	12,615,789	10,977,615	Random Forest +NAM	8,143,458	1.549
5	11,670,330	10,914,924	FICO-only +NAM	8,114,952	1.438
6	11,940,761	11,670,548	FICO-only +NAM	8,324,896	1.434
7	11,195,960	10,646,014	FICO-only +NAM	8,141,550	1.375
8	11,791,786	11,916,461	FICO-only +NAM	7,965,561	1.480
9	11,971,662	10,515,902	Random Forest +NAM	8,217,906	1.457
10	12,051,848	9,824,572	Random Forest +NAM	8,352,021	1.443
11	12,076,407	12,163,059	FICO-only +NAM	7,973,983	1.514
12	11,596,799	10,390,463	FICO-only +NAM	8,188,814	1.416
13	12,062,611	10,493,538	FICO-only +NAM	8,318,120	1.450
14	12,556,453	10,888,336	Random Forest +NAM	8,531,416	1.472

Seed	Engine expected ($)	Engine pseudo-real ($)	Best passive regime	Best passive ($)	Ratio
15	11,760,923	9,975,440	Random Forest +NAM	7,897,288	1.489
16	11,839,474	10,913,800	FICO-only +NAM	7,756,807	1.526
17	12,226,060	11,167,949	FICO-only +NAM	8,333,860	1.467
18	11,720,891	11,370,991	FICO-only +NAM	7,823,346	1.498
19	12,397,560	10,073,031	FICO-only +NAM	8,410,107	1.474

All models re-estimated on each training fold; analytic expected profit, no Monte-Carlo step. Realized net profit is positive for every regime in every split.

Appendix B. Base-Case Parameters

Group	Parameter	Value	Source
Horizon and discounting	n, r_{disc}	60 months, 5% p.a.	assumption
Loss profile [F1]	$\bar{\tau}$, LGD	12 months, 0.8	anchored to unsecured-consumer statistics; grid in Table 9
PD [F2]	source	Random Forest (500 trees), Platt-scaled on 5-fold out-of-fold train predictions	estimated
Stiglitz-Weiss response	γ_{emp}, δ_{emp}	0.089788, 6.574745	estimated (Table 1)
Acquisition cost	κ, ϕ, γ_{cac}	0.02, 0.20, 0.15	calibrated
Demand [F3]	α, λ, η_0, ξ, η_{cap}, η_{cut}	1.0, 0.003, 0.05, 0.5, 4, 0.02	calibrated; breakeven in Table 11
Types [F6]	m_k, π_k, attrition ω	{0.5, 1.5}, {0.5, 0.5}, 0	assumption; Table 12
Capital [F5]	h, ρ	10%, Basel other-retail $\rho(PD)$ (0.03–0.16)	BCBS (2005); Basel Framework CRE31
Constraints [F7]	max hike, max rate multiple, max PTI	+5 pts, 1.5×, max(0.15, original)	assumption
Gate [F4]	rule	mechanism value ≥ CAC	–
Grid	principal × term × rate shift	19 × 2 × 31 = 1,178	–
Simulation	N_{MC}, VaR level, ρ_{EC}	10,000, 99.9%, Basel other-retail (stress: 0.15)	–
Seeds	split / RF / out-of-fold / Z / MC	42 / 123 / 124 / 101 / 42	–

Appendix C. Notation

Symbol	Meaning
L, R, T	principal (\$), annual contract rate (%), term (months); subscript $orig$ or 0 for the original contract, c for a candidate
PMT, B_t	constant monthly French payment; outstanding balance after t payments
r_{disc}, r_m, v	annual cost of funds (5%), monthly rate $r_{disc}/12$, monthly discount factor $(1 + r_m)^{-1}$
$a(k, r)$	present value of \$1 per month for k months at rate r
$\bar{\tau}$, LGD	months to default; loss given default on the balance outstanding at default
PV_{nd}, PV_d	present value of the contract's cash flows conditional on no default / on default
NPV_{cond}, NPV_{real}	expected NPV under the model PD; NPV under the observed default outcome
EAD	exposure at default, $LGD \cdot B_{\bar{\tau}-1}$
PD_{base}, PD_{dyn}	calibrated PD of the original contract; PD of a candidate after the Stiglitz–Weiss shock
$\gamma_{emp}, \delta_{emp}$	estimated semi-elasticities of default log-odds to the rate and to payment-to-income
PTI, Y	payment-to-income ratio; annual income
α, λ	acceptance elasticities to principal cuts and to payment increases
$\eta_0, \eta_i, \xi, \eta_{cut}$	base hike aversion; borrower-specific hike aversion; its heterogeneity exponent; attraction of rate cuts
k, m_k, π_k	borrower type (low / high sensitivity), its sensitivity multiplier, its prior probability
$p_k, \bar{p}$	type-k and prior-weighted acceptance probabilities
$\kappa, \phi, \gamma_{cac}$	acquisition-cost base rate, size elasticity, rate elasticity
h, $K(PD)$, $\rho(PD)$	hurdle rate on capital; Basel ASRF capital requirement per \$ of EAD; asset correlation
X	discounted exposure duration per \$ of principal (years), the weight of the capital charge
C^{cap}	capital charge of a contract, $h\,K\,L\,X$
P_1, P_2, P_3, P_w	probabilities of the Tier-1, Tier-2, Tier-3 and walk-away routes
ω	attrition after a rejected Tier-1 offer
$\mathcal{V}_i$	mechanism value of borrower i (expected capital-adjusted value of the tree)
$\varepsilon_V, \varepsilon_p$	semi-elasticities of contract value and of acceptance with respect to the rate (Proposition 1)
Z, N_{MC}	Vasicek systematic factor; number of Monte-Carlo replications
$EL, VaR_{99.9}$, $CVaR_{99.9}, EC_{99.9}$	expected loss, 99.9% value-at-risk, conditional value-at-risk, economic capital
a_i, Π, p_i^*	passive approval indicator, passive profit, breakeven PD of borrower i (Section 1.2)
ℓ	out-of-sample log-likelihood of a PD model (Section 5.3)

Appendix D. Proof of Proposition 1

Fix $L = L_0$ and $T = T_0$ and consider $R \geq R_0$. With principal unchanged the size term of the acceptance function equals one, and each type's acceptance is $p_k(R) = s(R)\, e^{-\eta_i m_k (R - R_0)}$ with $s(R) = e^{-\lambda[PMT(R) - PMT_0]}$, so that $\bar{p}(R) = s(R) \sum_k \pi_k e^{-\eta_i m_k (R-R_0)}$ and

$$-\frac{\bar{p}'(R)}{\bar{p}(R)} = \lambda\, PMT'(R) + \eta_i \frac{\sum_k \pi_k m_k e^{-\eta_i m_k (R-R_0)}}{\sum_k \pi_k e^{-\eta_i m_k (R-R_0)}} = \lambda\, PMT'(R) + \eta_i \sum_k \omega_k(R)\, m_k = \varepsilon_p(R).$$

Where $V(R) > 0$ the objective $W = \bar{p}\,V$ is positive and $\ln W = \ln \bar{p} + \ln V$, so $d \ln W / dR = \varepsilon_V - \varepsilon_p$, which has the sign of W'.

(i) At R_0 the weights equal the priors, $\omega_k(R_0) = \pi_k$, and the multipliers are mean-preserving, $\sum_k \pi_k m_k = 1$, so $\varepsilon_p(R_0) = \lambda\, PMT'(R_0) + \eta_i$. A marginal hike raises W if and only if $\varepsilon_V(R_0) > \varepsilon_p(R_0)$.

(ii) An interior maximizer of W is an interior maximizer of $\ln W$ and satisfies the first-order condition $\varepsilon_V(R^*) = \varepsilon_p(R^*)$.

(iii) If $\varepsilon_V - \varepsilon_p$ crosses zero at most once, and from above, then $\ln W$ is increasing and then decreasing (or monotone) on $[R_0, R_0 + \bar{\Delta}]$, that is, W is quasi-concave there. If the difference is non-positive at R_0 it cannot become positive later, so W is non-increasing and the optimum is R_0; if it is non-negative at $R_0 + \bar{\Delta}$ it was positive throughout, so W is non-decreasing and the optimum is the cap; otherwise the single root is the unique maximizer.

(iv) Differentiating $V = -L + (1 - PD)\, PV_{nd} + PD\, PV_d - C^{cap}$ gives $V' = (1 - PD)\, PV'_{nd} + PD\, PV'_d - (PV_{nd} - PV_d)\, PD' - C^{cap\,\prime}$, and differentiating the logit of Section 4.4 gives $PD' = PD(1 - PD)\,[\gamma_{emp} + \delta_{emp} PMT'(R)/(Y/12)]$. At R_0 neither $V(R_0)$ nor $\varepsilon_p(R_0)$ depends on γ_{emp} or δ_{emp}, while $V'(R_0)$ is strictly decreasing in both: the direct term because $PV_{nd} - PV_d > 0$ (default destroys value), and the capital term because $C^{cap\,\prime} = hL[K'(PD)\, PD'\, X + K\, X']$ increases with PD' wherever K is increasing in PD, which with $LGD = 0.8$ and the other-retail correlation holds for PD below about 0.40, a range that covers more than 99% of the test population. Hence $\varepsilon_V(R_0)$ falls and, by (i) and (iii), the set of borrowers for whom a hike is optimal shrinks as γ_{emp} or δ_{emp} rises; it shrinks as η_i rises because $\varepsilon_p(R_0)$ is increasing in η_i. Finally, $\ln \sum_k \pi_k e^{-\eta_i m_k (R - R_0)}$ is a log-sum-exp of affine functions of R and therefore convex, so the type-mixture component of ε_p is non-increasing in R: as the rate rises the price-sensitive type selects out first. □

Appendix E. Reproducibility

Every number in this thesis is produced by one script, `sidur_13.R` (R 4.5.1 on macOS 15.6, aarch64; packages `randomForest` 4.7-1.2, `pROC` 1.19.0.1, `dplyr` 1.2.0), run from the thesis folder on the file `data_for_final.csv`. The random seeds are fixed (split 42, Random Forest 123, out-of-fold folds 124, systematic factor 101, Monte-Carlo draws 42, bootstrap 2026), so a re-run reproduces every table to the dollar; the addendum of Sections 5.2, 5.3, 5.5 and 5.10 begins with an automatic check that the base-case numbers of the 2026-09-15 freeze are reproduced before any new quantity is computed. The Python package `engine13/` is an independent reference implementation of the same engine, used for development, for the feature decomposition of Table 15 and for Figure 9; because Random Forest estimates are not bit-identical across software, the Python and R base cases agree on the engine-to-passive ratio (1.440 in both, before sub-grade pooling) rather than on every dollar. Figures 1–8 and 11 are drawn by `figs13.py` directly from the R output tables, Figure 9 by `fig9_geometry.py`, and Figure 10 is the R histogram in `sidur_13_plots.pdf`.

Tables	Output file(s) of `sidur_13.R`
1	Empirical_StiglitzWeiss_Regression_v13.csv
1b	Calibration_Diagnostics_v13.csv
2a, 2b, 4, 8	Master_Summary_Matrix_v13.csv (and the _rho015 variant)
3	Bootstrap_Inference_v13.csv
4, 4b	Model_AUC_Comparison_v13.csv, DeLong_AUC_Tests_v13.csv
4c, 4d	Likelihood_PD_Models_v13.csv, Vuong_Tests_v13.csv
5	Engine_Contract_Distribution_v13.csv
6	Heterogeneity_v13.csv
7	Axis_Decomposition_v13.csv
7b	Optimality_Gap_v13.csv
9	Scenario_Grid_v13.csv, Master_Summary_Matrix_v13_conservative.csv
10	Engine_Sensitivity_Tornado_v13.csv
11, 12	Breakeven_Elasticity_v13.csv, Sensitivity_Extra_v13.csv
13	ZeroCAC_v13.csv, PD_Source_Robustness_v13.csv, Identification_Robustness_v13.csv, Empirical_StiglitzWeiss_FE_v13.csv
13b	LR_Tests_StiglitzWeiss_v13.csv
14, Appendix A	Repeated_Splits_v13.csv, Repeated_Splits_Summary_v13.csv
15	engine13/out/final/fix_ladder.csv (Python reference implementation)
Section 4.11	Master_Summary_Matrix_v13_frozenspec.csv

List of Figures

Figure 1. Net profit by underwriting regime (expected, realized, pseudo-realized). Figure 2. Discrimination versus value. Figure 3. Tornado. Figure 4. Breakeven demand elasticity. Figure 5. Loss-profile scenarios. Figure 6. Split stability across twenty partitions. Figure 7. Mechanism and deployability sensitivities. Figure 8. Return–capital frontier under the Basel and $\rho = 0.15$ correlations. Figure 9. Geometric interpretation of a loan contract and of the engine's search. Figure 10. Distribution of Tier-1 rate shifts, Group 1. Figure 11. Axis decomposition and heterogeneity by credit grade.

Manuscript generated against the official `sidur_13.R` runs of 2026-09-15 (freeze) and 2026-09-25 (addendum, which reproduced every table of the freeze byte for byte); R 4.5.1, seed 42, $N_{MC} = 10{,}000$. The correspondence between tables and output files is given in Appendix E.